

Comparing deformation subspaces across anatomical variability: a population finite-element study of the human pelvis

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Abstract

Comparing deformation modes across subject-specific finite-element models requires distinguishing changes in mechanical behaviour from changes in spectral ordering. We investigated this distinction in 278 computed-tomography-derived pelvic models with individual geometry and bone elasticity on a shared reference domain. The 15 lowest linearised stiffness modes were compared using spectral gaps, modal assignment, principal angles, anatomical diameter functionals, and modal strain-energy fractions under standing and internal-ring expansion loads. Modes 1–3 formed a well-separated, order-stable group with label-exchange rates below 0.4%. A recurrent rank 9–10 cluster occurred in 54.7% of subjects. Within this cluster, median anteroposterior inlet coupling assigned to rank 9 decreased from 0.454 to 0.139 mm/mm between subjects without and with label exchange, whereas the corresponding subspace capacity changed from 0.551 to 0.525 mm/mm. Thus, the approximately 69% difference at one rank greatly exceeded the approximately 5% difference at the subspace level. Standing primarily recruited low-rank modes, whereas internal-ring expansion preferentially recruited the 9–10 block. These findings favour comparing compact modal subspaces alongside individual modes in crowded spectral regions. The conclusions concern initial elastic deformation under the specified pelvic loads.

Keywords: Population biomechanics, Pelvic mechanics, Finite-element modelling, Modal analysis, Anatomical variability, Invariant subspaces

1 Introduction

The human pelvis transfers loads between the trunk and lower limbs and surrounds a canal whose dimensions have anatomical and obstetric significance. Its deformation depends on bone geometry and material properties, joint compliance, and ligamentous restraints (Snijders et al. 1993; Vleeming et al. 2012). Pelves differ in shape, but it remains unclear whether that variation also changes the deformation pathways available to the ring and how different loads recruit them. We focus on the initial structural response under small deformations.

Population comparisons require multiple anatomies because average model inputs can produce responses that differ from the average response (Cook and Robertson 2016). Population modelling therefore has a role in computational biomechanics (Fernandez et al. 2020). Statistical pelvic shape models rank geometric variance (Arand et al. 2019); FE deformation modes rank mechanical response. Mechanical comparisons also require material properties, joint and ligament representations, and boundary conditions. CT-derived FE studies have examined strain variation during gait in five specimen-specific pelvis (Salo et al. 2017) and the effects of pelvic ligaments (Henyš et al. 2022). A separate statistical-shape-based musculoskeletal model estimated pelvic joint reactions during standing and walking in female participants (Chen et al. 2026). These studies characterise anatomy- or load-dependent responses. Our question is how to compare the underlying deformation directions across a larger population.

A stiffness eigenproblem provides an ordered set of deformation directions independent of any one prescribed load. Pelvic modal analysis is established: Henyš and Čapek (2019) studied convergence and experimental validation of a CT-derived pelvic-bone model, while Henyš et al. (2021) related stiffness-spectrum descriptors to anthropometry and sex. Across subjects, the same eigenvalue rank can represent different anatomical deformations. Near neighbouring eigenvalues, mode shapes can exchange their order or mix as parameters change, as recognised in structural mechanics (Pierre 1988; Manconi and Mace 2017). Invariant-subspace methods address this ambiguity by comparing the span of neighbouring modes (Ghosh and Ghanem 2012); the question here is whether that distinction changes the interpretation of pelvic deformation.

Suppose an eigenvector opens the inlet in one pelvis, while its neighbouring eigenvector carries most of that response in another. A rank-by-rank comparison would suggest a large change even if the two-mode space retains a similar inlet-opening capacity. We therefore measure spectral gaps, cross-subject correspondence, and anatomically defined diameter responses together. Basis-invariant capacity separates internal relabelling from changes in the subspace response.

Inlet and outlet diameter functionals make these comparisons anatomically interpretable. The maximum diameter change attainable within a modal subspace, at a fixed maximum nodal displacement, measures its kinematic capacity. It is distinct from compliance under an applied force. Static modal strain-energy fractions assess which directions the prescribed loads actually recruit. Considering capacity and recruitment together prevents an available deformation direction from being mistaken for one that a given load strongly excites.

We analyse 278 CT-derived pelvic FE models with subject-specific geometry and bone elasticity on a shared reference domain (Henyš and Kuchař 2025). We ask whether low-rank modes retain correspondence, whether recurrent clustered subspaces show smaller differences in anatomical capacity than individual ranks across label-exchange groups, and whether standing and internal-ring expansion recruit different spectral regions. The contribution is a population-level assessment of the appropriate modal comparison unit for anatomical interpretation. The rank groups and load responses reported below apply to this linearised model family.

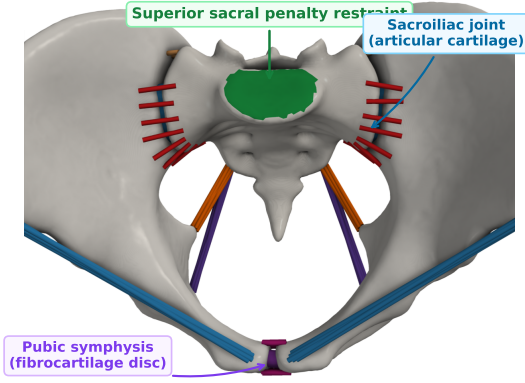
2 Materials and methods

We first define the shared-domain FE models and modal comparisons, then the anatomical functionals and load analyses used to interpret them. We distinguish solver rank j , reference label $i = \pi_s^{-1}(j)$ assigned across subjects, and anatomical functional ℓ throughout.

2.1 Cohort and common-reference geometry

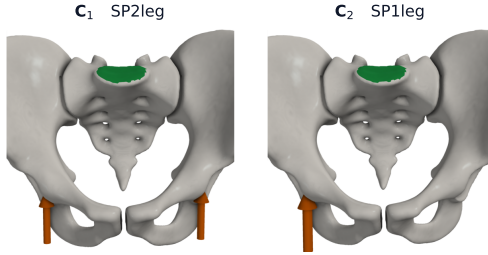
We analysed a precomputed set of $N=278$ CT-derived pelvic finite-element models assembled on a common reference domain ($n_F=150$, $n_M=128$; age 16–91 yr, mean 54 ± 19). The analysis reuses the published BoneDat cohort. The source repository provides standardised pelvic geometries suitable for in silico analysis (Henyš and Kuchař 2025). The reference pelvic domain Ω_{ref} was discretized using a conforming volumetric mesh comprising 166 808 linear four-node tetrahedral elements (Lagrange P_1 , 48 291 nodes, totaling $n = 144 873$ displacement degrees of freedom), with conforming interfaces between the bony segments (sacrum, left and right ossa coxae) and the joint cartilage domains (bilateral sacroiliac articular cartilage and the pubic symphysis disc). A nonrigid volumetric registration $\chi_s : \Omega_{\text{ref}} \rightarrow \Omega_s$ with deformation gradient $F_s(X) = \nabla_X \chi_s(X)$ and Jacobian $J_s(X) = \det F_s(X)$ maps the template domain Ω_{ref} onto each subject domain Ω_s . The registration was obtained by radial-basis-function interpolation. The registered geometries share anatomical correspondence and retain cohort size variation. The primary analysis uses archived solver-order spectra; the allometric analysis examines isotropic size

A Pelvic finite-element assembly & soft tissues



- Anterior SIJ lig.
- Interosseous SIJ lig.
- Sacrospinous lig.
- Posterior SIJ lig.
- Sacrotuberous lig.
- Pubic ligaments

Standardised standing loads



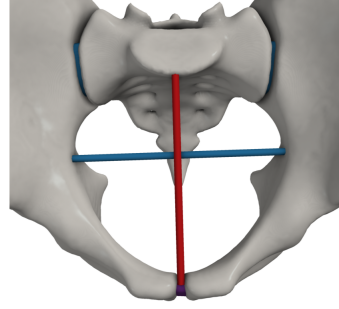
Bilateral standing

2 × 400 N vertical
Acetabular notches

Unilateral standing

1 × 800 N vertical
Right acetabular notch

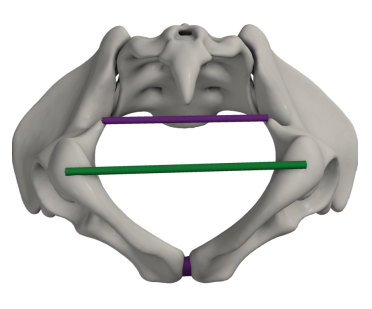
B₁ Inlet plane (superior view)



Pelvic Inlet Diameters

- Conjugata vera (AP): 123 mm
- Transverse diameter (ML): 131 mm

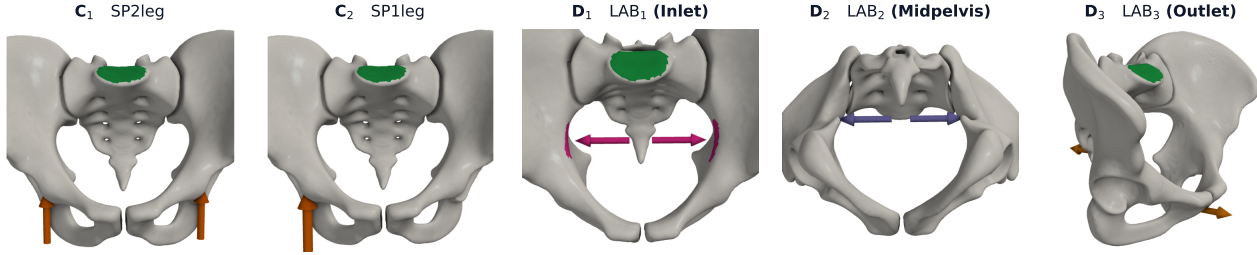
B₂ Outlet plane (inferior view)



Pelvic Outlet Diameters

- Interspinous diameter (BIS): 103 mm
- Bituberous diameter (BIT): 132 mm

Comparative pelvic expansion loads



Inlet ring expansion

±400 N lateral
Iliopectineal inner ring

Tuberosity distraction

±400 N lateral
Ischial tuberosities

AP outlet distraction

±400 N anteroposterior
Pubic crest & SCJ

Fig. 1 Pelvic model and standardised loading. (A) Bony segments, joint cartilage, ligament bundles, and superior sacral restraint. (B) Inlet and outlet diameter definitions on the reference anatomy. (C) Two-leg and one-leg standing loads. (D) Comparative internal-ring expansion, ischial distraction, and anteroposterior outlet distraction loads (LAB₁–LAB₃). Each traction is normalised to the prescribed resultant across subjects.

as a covariate. Fig. 1 summarises the anatomical dimensions, ligament representation, and boundary-condition layout used to define the pelvic models analysed here.

2.2 Pullback-assembled linear elasticity on a common reference domain

Let $\Omega_{\text{ref}} \subset \mathbb{R}^3$ be the template domain and Ω_s the subject domain, related by a fixed volumetric registration $x = \chi_s(X)$ with deformation gradient $F_s(X) = \nabla_X \chi_s(X)$ and local Jacobian $J_s(X) = \det F_s$. Writing $\chi_s(X) = X + \varphi_s(X)$, so $F_s = I + \nabla_X \varphi_s$, small-strain linear elasticity is solved on Ω_s but integrated on Ω_{ref} through the standard change of variables (Marsden and Hughes 1983; Bonet and Wood 1997). For trial and test fields $\hat{\mathbf{u}}, \hat{\mathbf{v}} \in V \subset [H^1(\Omega_{\text{ref}})]^3$, the physical strain tensor at $x = \chi_s(X)$ is

$$\begin{aligned} \boldsymbol{\varepsilon}_x(\mathbf{u}) &= \text{sym}(\nabla_x \mathbf{u}) = \text{sym}(\nabla_X \hat{\mathbf{u}} F_s^{-1}), \\ \boldsymbol{\sigma}_x &= C_{\text{phys}}(x) : \boldsymbol{\varepsilon}_x(\mathbf{u}), \end{aligned} \quad (1)$$

and the pullback-assembled elastic bilinear form reads

$$a(\hat{\mathbf{u}}, \hat{\mathbf{v}}) = \int_{\Omega_{\text{ref}}} \left[C_{\text{phys}}(\chi_s(X)) : \text{sym}(\nabla_X \hat{\mathbf{u}} F_s^{-1}) \right] : \text{sym}(\nabla_X \hat{\mathbf{v}} F_s^{-1}) J_s \, dX. \quad (2)$$

The fixed image registration supplies F_s and J_s for the small-strain formulation. The volume-metric bilinear form (denoted by m and assembled as M_s) inherits the geometric volume factor,

$$m(\hat{\mathbf{u}}, \hat{\mathbf{v}}) = \int_{\Omega_{\text{ref}}} J_s(X) \hat{\mathbf{u}} \cdot \hat{\mathbf{v}} dX, \quad (3)$$

defining an L^2 volume inner product weighted by local volumetric dilation. The resulting generalised eigenvalues $\lambda_k = \phi_k^\top K_s \phi_k / (\phi_k^\top M_s \phi_k)$ are stiffness-spectrum descriptors in a volume metric. With coordinates in mm and modulus in MPa, they have units N/mm⁴. The geometric-volume mass form makes these static stiffness descriptors; their reciprocals have units mm⁴/N. The *unmapped* reference mass matrix M_{ref}^0 is obtained from (3) with $J \equiv 1$ and provides a shared, registration-independent inner product across subjects.

Material model. The physical elasticity tensor $C_{\text{phys}}(x)$ is isotropic and heterogeneous. For bone, the CT-derived hydroxyapatite density $\rho(x)$ is converted to ash density $\rho_{\text{ash}} = (\rho + 0.09)/1.14$ and mapped to a Young's modulus $E(x)$ using an implemented piecewise approximation motivated by Keller (Keller 1994),

$$E(x) = \begin{cases} 2398 \text{ MPa}, & \rho_{\text{ash}}(x) \leq 0.486, \\ \alpha \rho_{\text{ash}}(x)^\beta, & \rho_{\text{ash}}(x) > 0.486, \end{cases} \quad (4)$$

$\alpha = 10\,200, \beta = 2.0,$

with Poisson's ratio $\nu = 0.3$. Negative input densities are clipped to zero. The law is applied at source points, after which the resulting modulus is interpolated to FE cells using RBF interpolation (smoothing parameter 50, ten neighbours). Here ρ and ρ_{ash} denote numerical density values in g/cm³, and α has units of MPa; logarithms of density below refer to the dimensionless numerical value in those units. The constant low-density branch and exponent 2.0 define the implemented approximation to the cited regression. Density-modulus calibration varies by anatomical site (Morgan et al. 2003); the cohort uses one prescribed law. Cartilage (sacroiliac joints and pubic symphysis) is homogeneous linear elastic with $E = 1.0$ MPa and $\nu = 0.45$. The cartilage parameters are prescribed effective values; pelvic joint mechanics and bone adaptation provide context for this representation (Frost 1987; Huiskes et al. 1987; Snijders et al. 1993; Vleeming et al. 2012; Goode et al. 2008).

Ligaments in pullback coordinates. Ligament families are represented as polyline bundles mapped from template nodes X_i to physical points $x_i = \chi_s(X_i)$. For each segment (X_i, X_j) define the physical unit tangent $\mathbf{e} = (x_j - x_i)/\|x_j - x_i\|$ and length $L = \|x_j - x_i\|$. The total bundle stiffness and pretension are distributed across N_{seg} segments, yielding per-segment $k_{\text{seg}} = K_{\text{bundle}}/N_{\text{seg}}$ and $T_{\text{seg}} = T_{\text{bundle}}/N_{\text{seg}}$. Each segment contributes an axial stiffness block

$$W = k_{\text{seg}} (\mathbf{e}\mathbf{e}^\top), \quad K_{ij}^{\text{ax}} = \begin{bmatrix} +W & -W \\ -W & +W \end{bmatrix}, \quad (5)$$

and, in the presence of pretension $T_{\text{seg}} \geq 0$, a geometric (prestress) stiffness

$$W_g = \frac{T_{\text{seg}}}{L} (I - \mathbf{e}\mathbf{e}^\top), \quad K_{ij}^{\text{geo}} = \begin{bmatrix} W_g & -W_g \\ -W_g & W_g \end{bmatrix} \succeq 0, \quad (6)$$

together with paired nodal forces $\pm T_{\text{seg}} \mathbf{e}$ on the right-hand side. The assembled linear stiffness, including prescribed ligament prestress, is

$$K_s = K_{\text{elas},s} + K_{\text{lig},s} + K_s^{\text{geo}}(T) + K_{\Gamma,s}, \quad f = f_{\text{ext}} + f_{\text{pret}}(T), \quad (7)$$

where geometric stiffness $K_s^{\text{geo}}(T)$ incorporates the effect of prescribed ligament pretension T in a linearised configuration. The term $K_{\Gamma,s}$ implements Dirichlet penalties on the restrained superior sacral facet Γ_{ref} , $a_\Gamma(\hat{\mathbf{u}}, \hat{\mathbf{v}}) = \gamma \int_{\Gamma_{\text{ref}}} \hat{\mathbf{u}} \cdot \hat{\mathbf{v}} J_{\Gamma,s} dS_X$ with $J_{\Gamma,s} = J_s \|F_s^{-T} N\|$, using a fixed penalty parameter $\gamma = 10^6$ N/mm³.

Force-controlled loading. Neumann terms are pulled back to Γ_{ref} so that physical resultants are invariant under χ_s :

$$\int_{\Gamma_s} \mathbf{t}_{\text{phys}} \cdot \mathbf{v} dS_x = \int_{\Gamma_{\text{ref}}} J_{\Gamma,s}(X) \mathbf{t}_{\text{phys}}(\chi_s(X)) \cdot \hat{\mathbf{v}}(X) dS_X. \quad (8)$$

All load cases in this study are defined by a total force $\mathbf{F}$ on a tagged physical region; the applied traction is $\mathbf{t}_{\text{phys}} = \mathbf{F}/A_{\text{phys}}$ with $A_{\text{phys}} = \int_{\Gamma_{\text{ref}}} J_{\Gamma,s} dS_X$, so that $\int_{\Gamma_s} \mathbf{t}_{\text{phys}} dS_x = \mathbf{F}$ for every subject.

2.3 Standardised load cases

Loads were applied as force-controlled Neumann tractions via (8) so that every subject experienced the same integrated force vector regardless of physical surface area. Two standing cases were considered: symmetric two-leg stance (SP2leg; 400 N per acetabulum, +CC) and one-leg stance (SP1leg; 800 N on the stance acetabulum,

+CC). The 400 N per hip and 800 N one-leg magnitudes define standardised comparative loads; measured hip-contact forces provide physiological context (Bergmann et al. 2001, 2016). Three comparative pelvic expansion cases probed load-dependent routing: LAB₁ applied bilateral mediolateral internal ring expansion forces (± 400 N outward), LAB₂ applied bilateral ischial-tuberosity distraction (± 400 N), and LAB₃ applied anteroposterior outlet distraction between the pubis insertion ($-AP$, 400 N) and the sacrococcygeal joint ($+AP$, 400 N). The three cases quantify initial pelvic compliance under prescribed force directions. Static finite-element solutions solve the linear equilibrium equation $K_s \mathbf{u} = \mathbf{b}$ with

$$\mathbf{b} = \mathbf{f}_{\text{ext}} + \mathbf{f}_{\text{pret}} \quad (9)$$

for the primary total-response analysis, where $\mathbf{f}_{\text{pret}}$ represents baseline ligament pretension (20 N per ligament bundle) and $\mathbf{f}_{\text{ext}}$ represents external load. Primary cohort-wide modal strain-energy fractions (Section 3.5; Table 4) are computed directly from these total displacement fields $\mathbf{u}$. By linearity, total displacement decomposes as $\mathbf{u} = \mathbf{u}_{\text{pret}} + \delta \mathbf{u}_{\text{ext}}$, where $\delta \mathbf{u}_{\text{ext}} = K_s^{-1} \mathbf{f}_{\text{ext}}$ is the incremental external load response. Under external load scaling $\mathbf{f}_{\text{ext}} \rightarrow c \mathbf{f}_{\text{ext}}$, incremental displacements scale proportionally ($\delta \mathbf{u}_{\text{ext}} \rightarrow c \delta \mathbf{u}_{\text{ext}}$), such that modal expansion coefficients $q_i = \phi_i^\top M_s \delta \mathbf{u}_{\text{ext}}$ scale by c , modal energies $E_i = \frac{1}{2} \lambda_i q_i^2$ scale by c^2 , and the normalised modal strain-energy fractions $f_i = E_i / \sum_j E_j$ of the incremental response are strictly scale-invariant. To examine the effect of baseline pretension on routing, we compared archived total-response and incremental-response modal fractions in 18 subjects with complete stored incremental outputs. The top recruited mode agreed for all five loads in all 18 subjects. The maximum individual change in the 9–10 block fraction was 4.24 percentage points, while its mean change under LAB₁ was +3.48 percentage points. The smallest dominant set reaching 80% of retained energy had the same size in all 18 standing cases and in 14 of 18 LAB₁ cases. Dominant routing was similar in this 18-subject subset.

2.4 Generalised eigenproblem, rigid-mode filtering, and cross-subject pairing

For each subject s we assembled (K_s, M_s) on Ω_{ref} via (7) and (3) and solved the generalised eigenproblem (the subject index on λ_k and ϕ_k is suppressed)

$$K_s \phi_k = \lambda_k M_s \phi_k, \quad \phi_i^\top M_s \phi_j = \delta_{ij}, \quad (10)$$

using a Krylov–Schur solver (SLEPc) with shift-invert spectral transform and MUMPS direct factorisation, extracting the $N_{\text{ev}} = 15$ lowest non-rigid eigenpairs. Rigid modes (numerically $\lambda < 10^{-8}$) were filtered. For cross-subject comparison we used the mass-weighted modal assurance criterion on the shared reference metric,

$$\text{MAC}_{M_{\text{ref}}^0}(\mathbf{x}, \mathbf{y}) = \frac{|\mathbf{x}^\top M_{\text{ref}}^0 \mathbf{y}|^2}{(\mathbf{x}^\top M_{\text{ref}}^0 \mathbf{x})(\mathbf{y}^\top M_{\text{ref}}^0 \mathbf{y})} \in [0, 1], \quad (11)$$

and solved the Hungarian optimal-assignment problem on $-\text{MAC}_{M_{\text{ref}}^0}$ over the full $N_{\text{ev}} \times N_{\text{ev}}$ matrix (Kuhn 1955), yielding an assignment bijection $\pi_s : \{1, \dots, N_{\text{ev}}\} \rightarrow \{1, \dots, N_{\text{ev}}\}$, where $\pi_s(i_{\text{ref}}) = j_{\text{rank}}$ maps reference identifier i_{ref} to its rank position in subject s , and the inverse permutation $\pi_s^{-1}(j_{\text{rank}}) = i_{\text{ref}}$ assigns reference identity to each solver rank. Assigned pairs with $\text{MAC}_{M_{\text{ref}}^0} < 0.05$ were flagged as unpaired for reporting purposes. This flag indicates a low-confidence match; the assignment remains a bijection. Rank index j (solver order) is required for gap statistics and cluster detection; the inverse assignment $\pi_s^{-1}(j)$ maps each solver mode back to its reference identity and is used only for anatomical interpretation, preserving the distinction between rank order and pathway identity emphasised in the Introduction.

2.5 Spectral gap structure and clustered-regime detection

For each subject with ordered eigenvalues $\lambda_1 \leq \dots \leq \lambda_{N_{\text{ev}}}$, relative consecutive gaps are

$$\text{gap}_{\text{in}}(i) = \frac{\lambda_{i+1} - \lambda_i}{\lambda_i}, \quad i = 1, \dots, N_{\text{ev}} - 1. \quad (12)$$

A clustered regime $[i..j]$ is defined as the maximal contiguous rank block starting at i with internal gaps below threshold, $\text{gap}_{\text{in}}(k) < \varepsilon_{\text{in}}$ for $k = i, \dots, j - 1$. The threshold is set from data as the 25th percentile of the pooled gap_{in} distribution across the cohort, yielding $\varepsilon_{\text{in}} \approx 0.13$; sensitivity to the percentile choice is reported in Section 3.7. External separation is quantified by

$$\text{sep}_{\text{out}}([i..j]) = \min \left(\frac{\lambda_i - \lambda_{i-1}}{\lambda_i}, \frac{\lambda_{j+1} - \lambda_j}{\lambda_j} \right), \quad (13)$$

defined for interior clusters ($i > 1$ and $j < N_{\text{ev}}$) where both adjacent neighbours exist. For boundary blocks reaching modal truncation (such as the 12–15 cluster with $N_{\text{ev}} = 15$), the upper neighbour is uncomputed and two-sided separation is formally undefined. All adjacent relative gaps exceeded the operational multiplicity threshold $\varepsilon_{\text{deg}} = 10^{-6}$; none met $\text{gap}_{\text{in}} \leq \varepsilon_{\text{deg}}$. The regimes detected by (12) are operational small-gap clusters associated with elevated perturbation sensitivity (Pierre 1988; Manconi and Mace 2017).

2.6 Invariant subspaces: principal angles, Grassmann distance, and Procrustes alignment

For each clustered rank block of dimension r , the corresponding subject-specific subspace was represented by an M_{ref}^0 -orthonormal basis

$$Q_s \in \mathbb{R}^{n \times r}, \quad Q_s^\top M_{\text{ref}}^0 Q_s = I_r. \quad (14)$$

With a fixed reference basis Q_{ref} , define the overlap matrix

$$S = Q_{\text{ref}}^\top M_{\text{ref}}^0 Q_s \in \mathbb{R}^{r \times r}, \quad S = U \Sigma V^\top, \quad \cos \theta_k = \sigma_k(S), \quad (15)$$

with singular values σ_k yielding the principal angles $\theta_k \in [0, \pi/2]$ between the two subspaces (Björck and Golub 1973). The Grassmann distance is

$$d_G(Q_{\text{ref}}, Q_s) = \|\theta\|_2 = \left(\sum_{k=1}^r \theta_k^2 \right)^{1/2}. \quad (16)$$

When a stable basis representation inside a clustered subspace is required, we apply an orthogonal Procrustes rotation (Schönemann 1966),

$$\tilde{Q}_s = Q_s (V U^\top), \quad (17)$$

which minimises the M_{ref}^0 -weighted distance $\|Q_{\text{ref}} - \tilde{Q}_s R\|_{M_{\text{ref}}^0}$ over orthogonal matrices $R \in O(r)$ while preserving the subspace $\text{span}(Q_s)$.

We compare subspaces through basis-invariant measures and retain individual-rank couplings as the comparator for label exchange.

Degeneracy vs. near-degenerate mixing. Exact mathematical multiplicity requires $\lambda_i = \lambda_{i+1}$ (operationalized numerically as $\text{gap}_{\text{in}}(k) \leq \varepsilon_{\text{deg}}$ for all k in the block, with $\varepsilon_{\text{deg}} \ll \varepsilon_{\text{in}}$); individual eigenvectors inside a degenerate subspace are then not unique, and any unitary rotation inside $\text{span}(Q_s)$ is an equally valid basis. When $\varepsilon_{\text{deg}} < \text{gap}_{\text{in}}(k) < \varepsilon_{\text{in}}$ the eigenvalues are close but distinct; each eigenvector is unique up to sign, yet small perturbations can rotate the internal basis substantially while the spanned subspace can be less sensitive if its external separation is sufficiently large relative to the perturbation. Anatomical coupling is therefore evaluated directly with the functionals below.

2.7 Canonical anatomical functionals inside clustered subspaces

To interpret clustered subspaces biologically, we use basis-invariant linear functionals defined by anatomical landmark pairs. On the reference template mesh, five key obstetric and locomotor diameters are defined:

- (i) **AP inlet** (s_{AP}): Promontory (X_{prom}) to superior pubic symphysis (X_{pub}), with unit direction $\mathbf{e}_{\text{AP}} = (X_{\text{pub}} - X_{\text{prom}}) / \|X_{\text{pub}} - X_{\text{prom}}\|$;
- (ii) **ML inlet** (s_{ML}): Widest transverse diameter across left and right iliopectineal lines ($X_{\text{left}}, X_{\text{right}}$), with unit direction $\mathbf{e}_{\text{ML}} = (X_{\text{right}} - X_{\text{left}}) / \|X_{\text{right}} - X_{\text{left}}\|$;
- (iii) **Interspinous midpelvic diameter (BIS)** (s_{BIS}): Left to right ischial spine ($X_{\text{sp, left}}, X_{\text{sp, right}}$), with unit direction $\mathbf{e}_{\text{BIS}} = (X_{\text{sp, right}} - X_{\text{sp, left}}) / \|X_{\text{sp, right}} - X_{\text{sp, left}}\|$;
- (iv) **Bituberous outlet** (s_{BIT}): Left to right inferior ischial tuberosity ($X_{\text{tub, left}}, X_{\text{tub, right}}$), with unit direction $\mathbf{e}_{\text{BIT}} = (X_{\text{tub, right}} - X_{\text{tub, left}}) / \|X_{\text{tub, right}} - X_{\text{tub, left}}\|$;
- (v) **AP outlet** (s_{OUTLETAP}): Sacrococcygeal joint (X_{cocc}) to inferior pubic symphysis ($X_{\text{pub, inf}}$), with unit direction $\mathbf{e}_{\text{OUTAP}} = (X_{\text{pub, inf}} - X_{\text{cocc}}) / \|X_{\text{pub, inf}} - X_{\text{cocc}}\|$.

For each subject s , $+$ denotes the target landmark and $-$ the origin. Their physical positions are $x_{\ell, \pm}^{(s)} = \chi_s(X_{\ell, \pm})$, and $\mathbf{e}_{\ell}^{(s)} = (x_{\ell, +}^{(s)} - x_{\ell, -}^{(s)}) / \|x_{\ell, +}^{(s)} - x_{\ell, -}^{(s)}\|$. The modal displacement degrees of freedom $\mathbf{u} \in \mathbb{R}^n$ remain indexed on the reference mesh. Let $\mathcal{I}_{\ell, \pm} \mathbf{u}$ denote inverse-distance-squared interpolation from the 32 nearest reference nodes to the corresponding template landmark $X_{\ell, \pm}$. The implemented functional is

$$s_{\ell}^{(s)}(\mathbf{u}) = \mathbf{e}_{\ell}^{(s)\top} (\mathcal{I}_{\ell, +} \mathbf{u} - \mathcal{I}_{\ell, -} \mathbf{u}) = \mathbf{b}_{\ell}^{(s)\top} \mathbf{u}, \quad \mathbf{b}_{\ell}^{(s)} \in \mathbb{R}^n, \quad \ell \in \{\text{AP, ML, BIS, BIT, OUTLETAP}\}. \quad (18)$$

The interpolation weights are fixed across subjects; the direction $\mathbf{e}_\ell^{(s)}$ is updated from RBF-mapped landmark positions. BIS measures the midpelvis between the ischial spines, whereas BIT measures the intertuberosus outlet. The AP-outlet observable runs from the sacrococcygeal landmark to the inferior pubic symphysis.

Basis-invariant max-nodal anatomical capacity. For any individual functional $\mathbf{b}_\ell^{(s)}$ and an orthonormal basis $Q_s \in \mathbb{R}^{n \times r}$ spanning a clustered subspace of dimension r ($Q_s^\top M_{\text{ref}}^0 Q_s = I_r$), every displacement in the subspace is parameterised by $\mathbf{u} = Q_s \boldsymbol{\alpha}$ with $\boldsymbol{\alpha} \in \mathbb{R}^r$. The maximum anatomical diameter response per 1 mm maximum nodal displacement is defined by the constrained max-nodal capacity:

$$\kappa_{\ell,\infty}(Q_s) = \max_{\boldsymbol{\alpha} \neq \mathbf{0}} \frac{|\mathbf{b}_\ell^{(s)\top} Q_s \boldsymbol{\alpha}|}{\max_a \|(Q_s \boldsymbol{\alpha})_a\|_2} = \max_{\boldsymbol{\alpha}} |\mathbf{b}_\ell^{(s)\top} Q_s \boldsymbol{\alpha}| \quad \text{s.t.} \quad \max_a \|(Q_s \boldsymbol{\alpha})_a\|_2 \leq 1, \quad (19)$$

where a indexes the finite-element mesh nodes and $\|(Q_s \boldsymbol{\alpha})_a\|_2$ denotes the Euclidean displacement norm of node a . Because both the numerator and the max-nodal constraint depend only on the represented displacement field $\mathbf{u} = Q_s \boldsymbol{\alpha}$, this quantity is strictly invariant under arbitrary internal orthogonal basis rotations $\tilde{Q}_s = Q_s R$ with $R \in O(r)$, since $\tilde{Q}_s(R^\top \boldsymbol{\alpha}) = Q_s \boldsymbol{\alpha}$. For $r = 2$, homogeneity allows parameterisation of $\boldsymbol{\alpha}$ by polar angle $\theta \in [0, \pi)$ with $\boldsymbol{\alpha}(\theta) = (\cos \theta, \sin \theta)^\top$; the maximum is approximated from 90 angular samples followed by scalar refinement. For $r \geq 3$, we solve the equivalent convex program $\max_{\boldsymbol{\alpha}} \mathbf{c}^\top \boldsymbol{\alpha}$ subject to $\|(Q_s \boldsymbol{\alpha})_a\|_2 \leq 1$ at every node, where $\mathbf{c} = Q_s^\top \mathbf{b}_\ell^{(s)}$; the sign symmetry removes the absolute value. Constraint generation starts with 16 nodes, solves each reduced problem with SLSQP and analytic Jacobians, adds the most violated mesh node, and terminates only when the full-mesh maximum squared norm is at most $1 + 10^{-8}$. The objective is linear and every constraint is convex, so a feasible stationary solution is a global optimum of the finite-dimensional problem up to numerical tolerance. Four independent full-constraint optimisations provide an additional check. The optimal displacement field $\boldsymbol{\psi}_\ell^{(s)} = Q_s \boldsymbol{\alpha}^* / \max_a \|(Q_s \boldsymbol{\alpha}^*)_a\|_2$ is normalised so that $\max_a \|\boldsymbol{\psi}_{\ell,a}^{(s)}\|_2 = 1$ numerically, and the resulting anatomical capacity $\kappa_{\ell,\infty}(Q_s) = |\mathbf{b}_\ell^{(s)\top} \boldsymbol{\psi}_\ell^{(s)}|$ has physical units of millimetres diameter change per millimetre maximum nodal displacement.

Multivariate canonical decomposition. When joint coupling across a pair of measurement vectors ($\mathbf{b}_1^{(s)}, \mathbf{b}_2^{(s)}$) (e.g., AP and ML inlet, or BIS and BIT outlet) is examined, we assemble the covector matrix $B_s = [\mathbf{b}_1^{(s)} \ \mathbf{b}_2^{(s)}] \in \mathbb{R}^{n \times 2}$ and its subspace projection:

$$C_s = Q_s^\top B_s = [Q_s^\top \mathbf{b}_1^{(s)} \quad Q_s^\top \mathbf{b}_2^{(s)}] \in \mathbb{R}^{r \times 2}, \quad (20)$$

$$C_s = U_s \Sigma_s V_s^\top,$$

where the subject-specific singular values $\sigma_{s,1} \geq \sigma_{s,2}$ describe joint coupling under the coefficient constraint $\|\boldsymbol{\alpha}\|_2 = 1$, equivalent to unit norm in M_{ref}^0 . Max-nodal capacities are evaluated separately by Eq. (19). Let $\mathbf{u}_{s,k}$ and $\mathbf{v}_{s,k}$ be column k of U_s and V_s , respectively. The joint-SVD displacement directions are $\boldsymbol{\zeta}_k^{(s)} = Q_s \mathbf{u}_{s,k}$ ($k = 1, 2$), with corresponding dual functionals $\tilde{\mathbf{b}}_k^{(s)} = B_s \mathbf{v}_{s,k} = \sum_{\ell=1}^2 (V_s)_{\ell k} \mathbf{b}_\ell^{(s)}$. Their action is $\tilde{\mathbf{b}}_k^{(s)\top} \boldsymbol{\zeta}_k^{(s)} = \sigma_{s,k}$. The singular-direction scores subsequently normalised by maximum nodal displacement are $\hat{s}_k^{(s)} = \sigma_{s,k} / \max_a \|\boldsymbol{\zeta}_{k,a}^{(s)}\|_2$. Because $V_s \in O(2)$ mixes the two anatomical axes, the first joint direction weights the first observable more strongly only when $|(V_s)_{11}| \geq |(V_s)_{21}|$. Across the pelvic cohort, the first joint direction in the 9–10 subspace weights the transverse (ML) observable more strongly in 99.6% of subjects. For axis-specific evaluation of anatomical opening, the basis-invariant single-functional capacity $\kappa_{\ell,\infty}(Q_s)$ (Eq. (19)) serves as our primary quantitative metric, while joint-SVD scores provide complementary bivariate structure.

The reference-mode displacement atlas and its polynomial decomposition are given in Appendix A.

2.8 Load routing and modal strain-energy fractions

For each load case we solved the static problem $K_s \mathbf{u}_s = \mathbf{b}$ on Ω_{ref} . With $\Phi_s = [\phi_1, \dots, \phi_{N_{\text{ev}}}]$ denoting the M_s -orthonormal retained modes, we projected the response as

$$\mathbf{q}_s = \Phi_s^\top M_s \mathbf{u}_s, \quad E_{s,i} = \frac{1}{2} \lambda_i q_{s,i}^2, \quad f_{s,i} = \frac{E_{s,i}}{\sum_{j=1}^{N_{\text{ev}}} E_{s,j}}. \quad (21)$$

The load-case index and the suppressed subject index on λ_i are implicit. We write f_i below when the subject and load are clear. The $E_{s,i}$ partition the quadratic energy of the retained projection under the full linearised K_s , including its ligament-geometric and boundary-penalty contributions; $f_{s,i}$ is normalised over the $N_{\text{ev}} = 15$ computed modes. Block throughput for a clustered regime C is $f_C = \sum_{i \in C} f_i$, and the dominant mode set N_{80} of a load case is the smallest subset of modes (selected in descending order of mean energy contribution) whose cumulative mean energy fractions reach at least 0.80.

First-order material-parameter sensitivity of eigenvalues and adjacent gaps is reported in Appendix D.

2.9 Decomposition of variability sources and robustness analyses

To separate shape and material contributions to spectral variability, we compared the full cohort against a shape-only cohort (χ_s , fixed reference bone modulus) and a material-only cohort (identity deformation, subject-specific bone modulus). Robustness was assessed by random permutation of subject order and 2,000 full-cohort bootstrap draws with replacement, re-estimating the cluster threshold in each draw. Exploratory allometric regressions and the variance decomposition of eigenvalues are reported in Appendix C.

2.10 Analytical interpretation of a close eigenvalue pair

For a symmetric two-coordinate stiffness matrix $K_2 = \begin{bmatrix} k_1 & c \\ c & k_2 \end{bmatrix}$, the eigenvalue separation is $2\sqrt{((k_1 - k_2)/2)^2 + c^2}$. Near $k_1 = k_2$, a small change in detuning can rotate the two eigenvectors while their joint span remains unchanged. Appendix E gives the full three-coordinate illustration and its stochastic implementation.

2.11 Statistical analysis

All statistical analyses were executed using SciPy, statsmodels, and custom Python routines. Continuous metrics across non-normal cohorts were evaluated using non-parametric methods. Two-group comparisons (e.g., non-swapped vs. swapped cluster members, or male vs. female distributions) were conducted using two-sided Mann–Whitney U tests. Effect sizes for Mann–Whitney tests are reported as rank-biserial correlations $r_{rb} = 1 - 2U_1/(n_1n_2)$, where U_1 is computed for the first listed group. With this orientation, positive values indicate larger observations in the second group; the statistic equals Cliff's delta for second versus first group. Bivariate associations between continuous anatomical dimensions (such as AP inlet diameter or isotropic scale) and modal metrics (energy fractions, spectral gaps) were quantified using Spearman's rank correlation coefficient ρ_s . The four rank-versus-capacity comparisons in Table 2 were adjusted with the Benjamini–Hochberg procedure. Exploratory sex-group tests are reported without multiplicity adjustment. Confidence intervals for median differences in the label-exchange comparisons were obtained by percentile bootstrap resampling (5,000 independent within-group resamples; random seed 20260918), using the 2.5th and 97.5th percentiles. Mann–Whitney probabilities compare distributions; bootstrap intervals quantify median differences. Cluster-prevalence intervals used 2,000 bootstrap replicates of 278 draws with replacement and threshold re-estimation (seed 20260918). These intervals describe resampling variability within the cohort.

Differences in subspace orientation across the cohort were quantified using the Grassmann distance $d_G(Q_{\text{ref}}, Q_s) = \|\theta\|_2$. The Grassmann distance d_G measures between-subspace orientation independently of internal basis indexing. Sex-group distances use the selected reference.

2.12 Software and computational implementation

Finite-element assembly, eigensolves, and postprocessing were implemented in Python 3.11 with DOLFINx/FEniCSx (≥ 0.10), PETSc, and SLEPc. Quadrature points and weights were obtained with Basix 0.10.0, the finite-element library used by FEniCSx. Study analysis scripts produced the figures, tables, and numerical results. Intermediate outputs are stored as compressed Zarr arrays.

3 Results

3.1 Population variability separates an order-locked backbone from recurrent clustered regimes

The cohort of $N=278$ subjects resolves the spectrum into two regimes: an order-locked low-rank backbone and recurrent higher-rank clustered blocks (Fig. 2, Table 1). The reference deformation atlas is provided in Appendix A. Modes 1–3 are rarely reordered across the cohort (swap rate $< 0.4\%$), whereas higher ranks contain small consecutive gaps and frequent label exchange. Using a data-driven cluster threshold $\varepsilon_{\text{in}} \approx 0.131$ (25th percentile of pooled relative consecutive gaps; see Methods), we detect nine near-degenerate rank clusters above a 3% prevalence floor. The most prevalent cluster is rank modes 9–10 (54.7% of subjects), followed by rank modes 12–15 (41.7%) and rank modes 5–6 (33.8%). All observed adjacent gaps exceeded the operational multiplicity threshold $\varepsilon_{\text{deg}} = 10^{-6}$.

The low-rank and higher-rank spectral regions show different exchange frequencies. Label fragility is concentrated in a small number of compact families, while low-rank order is preserved. The 9–10 and 12–15 blocks have small median internal gaps and show label exchange across the archived cohort. These are cross-sectional exchange observations. The 9–10 block has a larger median external separation than internal gap, whereas the 12–15 regime

Table 1 Recurrent clustered rank blocks above the 3% prevalence floor. External separation is reported as (—) for blocks reaching the 15-mode truncation boundary

Cluster	Dim	n	Prev.	Med. gap _{in,min}	Med. sep _{out}	Type
Rank modes 9–10	2	152	54.7%	0.08114	0.4926	near-degenerate
Rank modes 12–15	4	116	41.7%	0.03153	—	near-degenerate
Rank modes 5–6	2	94	33.8%	0.08737	0.2345	near-degenerate
Rank modes 12–13	2	78	28.1%	0.03839	0.1941	near-degenerate
Rank modes 14–15	2	70	25.2%	0.05116	—	near-degenerate
Rank modes 12–14	3	45	16.2%	0.04109	0.1855	near-degenerate
Rank modes 4–5	2	35	12.6%	0.09521	0.2185	near-degenerate
Rank modes 13–15	3	29	10.4%	0.04465	—	near-degenerate
Rank modes 7–8	2	21	7.6%	0.1051	0.3676	near-degenerate

is left-separated but right-truncated by the eigensolve; the 9–10 pair can therefore be assessed as an interior subspace, whereas conclusions about isolation of 12–15 remain limited. The result identifies spectral regions in which rank-based comparison is more or less reliable under the present model assumptions.

Per-mode pairing shows the highest swap rates at modes 12 and 13; mode 15 also has a sizeable fraction of matches flagged as unpaired under the specified MAC threshold (Fig. 2B). Modes 1–3 rarely change order. These observations motivate subspace comparisons for recurrent higher-rank blocks while retaining individual-mode comparisons at low rank.

A variability decomposition into shape-only and material-only datasets (Fig. 2C) shows that the shape-only dataset has the larger coefficient of variation for modes 1–8, whereas bone-density heterogeneity contributes with increasing relative weight in modes 9–15. In these two restricted cohorts, geometric variation contributes more to lower-rank eigenvalue variability, while material-only variability becomes relatively more prominent at higher ranks. The two restricted cohorts quantify separate coefficients of variation; interaction effects remain within the full-cohort comparison.

In 2,000 full-cohort bootstrap draws with threshold re-estimation, 95% percentile prevalence intervals were 49.6–59.4% for 9–10 and 36.7–46.0% for 12–15. Permuting subject order left cluster detection unchanged.

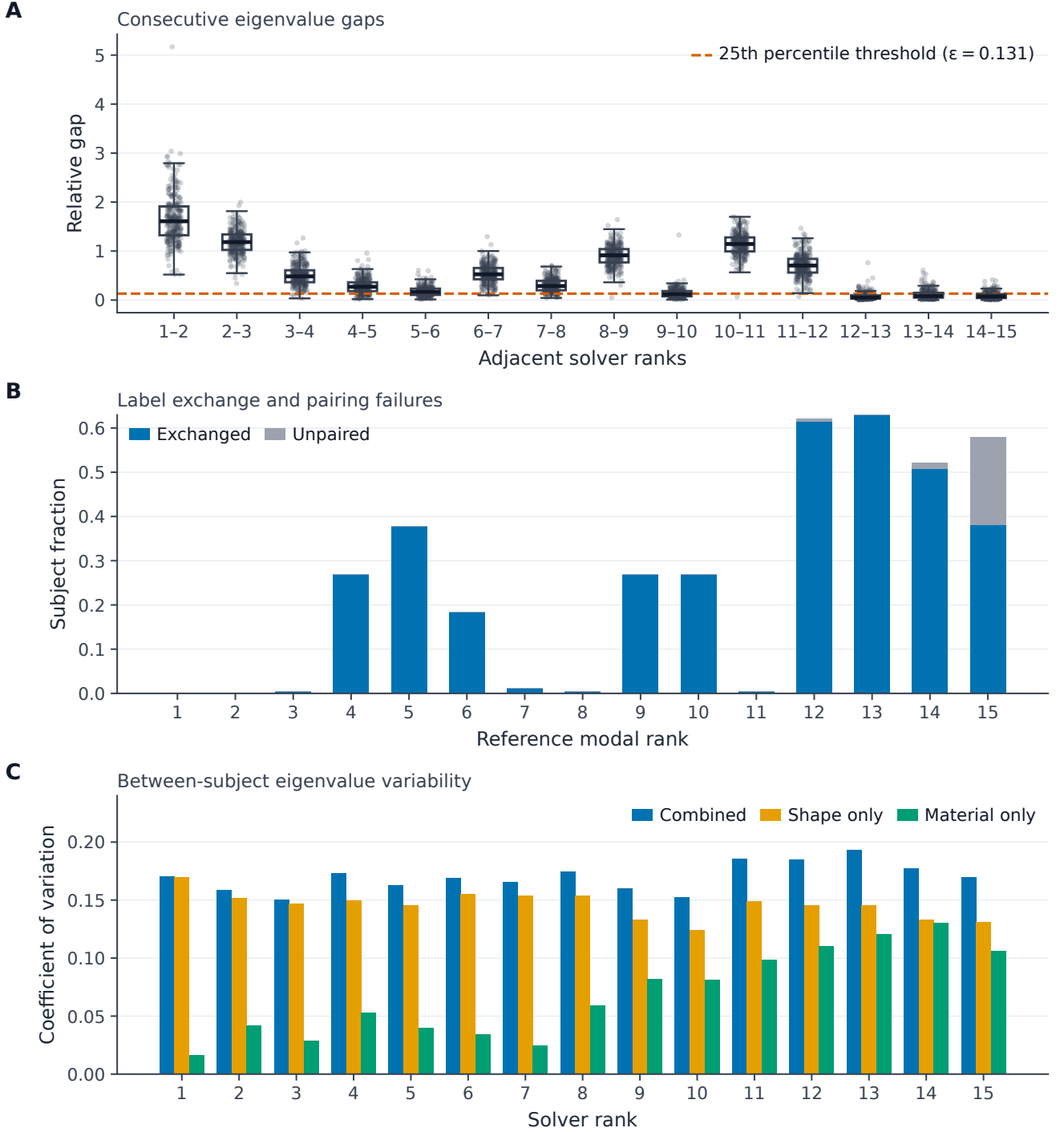


Fig. 2 Population variability partitions the stiffness spectrum into an order-locked backbone and clustered higher ranks. (A) Consecutive relative spectral gaps $\text{gap}_{\text{in}}(i) = (\lambda_{i+1} - \lambda_i)/\lambda_i$ across the cohort ($N=278$); the dashed line marks the data-driven cluster threshold $\epsilon_{\text{in}} \approx 0.131$ (25th percentile). (B) Subject fraction exhibiting label exchange (blue) and matches flagged as unpaired (grey; $\text{MAC}_{M_{\text{ref}}}^0 < 0.05$) per reference label. (C) Eigenvalue variability (coefficient of variation, CoV) across the combined cohort (blue), shape-only cohort (orange), and material-only cohort (green).

3.2 Anatomical coupling distinguishes recurrent subspaces

Because rank labels exchange, we next compare the spanned subspaces. Grassmann distance measures their orientation relative to one reference; basis-invariant diameter capacities quantify the AP and ML inlet, interspinous midpelvic, and bituberous outlet responses available within each block (Methods; Fig. 3).

The rank 9–10 subspace has median AP-inlet capacity 0.54 mm/mm and ML-inlet capacity 1.05 mm/mm. The 5–6 block also accommodates outlet deformation, with interspinous capacity 0.95 mm/mm, bituberous capacity 1.48 mm/mm, and modelled AP-outlet capacity 0.50 mm/mm. The corresponding 9–10 capacities are 1.20, 1.55, and 1.28 mm/mm, respectively. The 5–6 block is recruited under ischial distraction (Section 3.5).

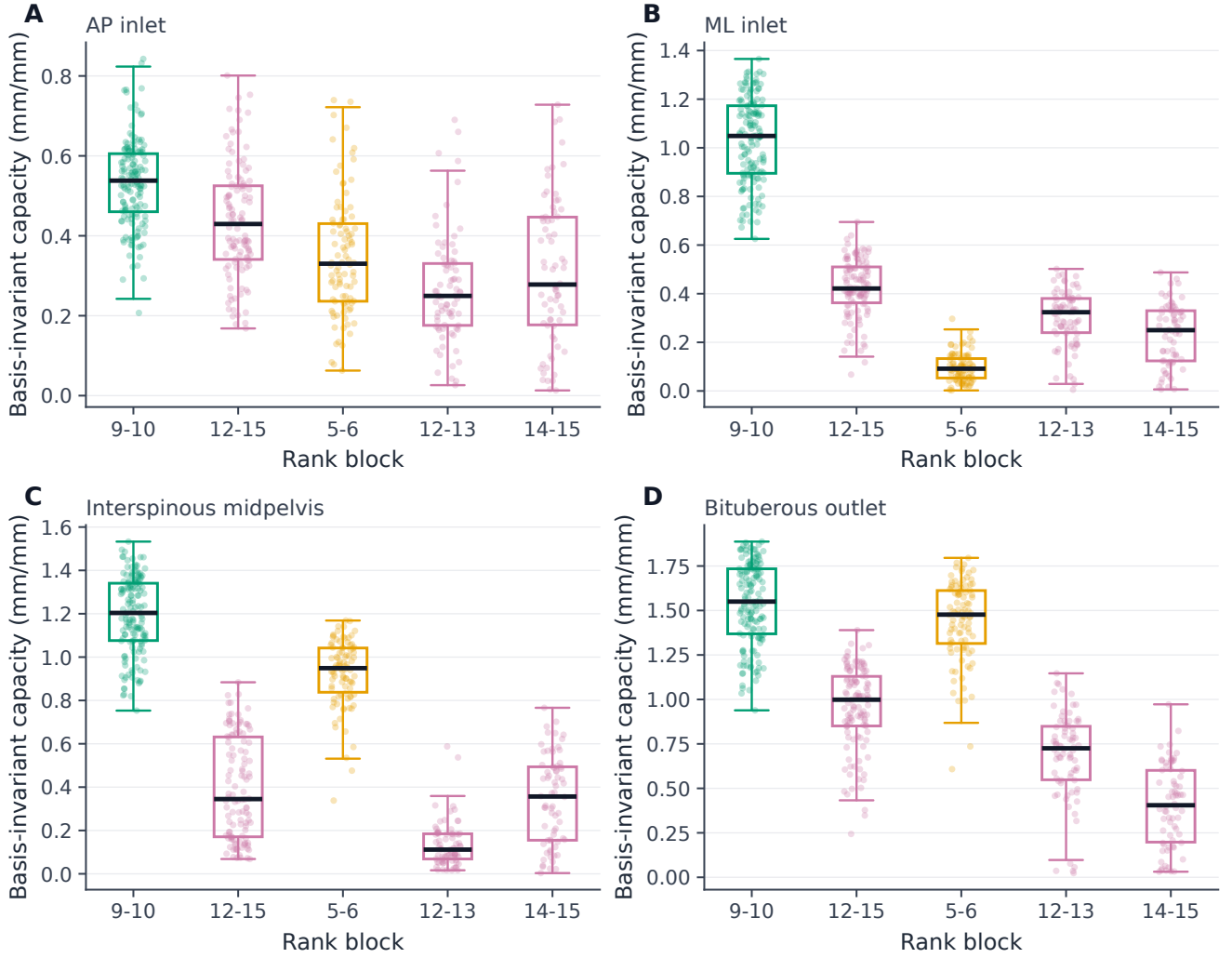


Fig. 3 Basis-invariant anatomical capacities across recurrent clustered modal subspaces. Basis-invariant anatomical capacities (maximum anatomical diameter response per 1 mm maximum nodal displacement) for the five most prevalent rank blocks across the cohort ($N=278$): (A) anteroposterior (AP) inlet, (B) mediolateral (ML) inlet, (C) interspinous midpelvic diameter, and (D) bituberous outlet. Boxes indicate median and interquartile range (IQR); whiskers extend to $1.5 \times \text{IQR}$.

The four-dimensional 12–15 block illustrates a third regime. Its capacities were evaluated using the constrained optimisation described in Methods. It is common and internally crowded, but its measured capacities are less specific to one diameter than those of the two-mode blocks. Because the eigensolve stops at mode 15, its upper external gap is unknown. We therefore describe 12–15 as a truncated higher-rank regime.

Exploratory sex-group comparisons of Grassmann distance show their largest rank-biserial contrast for the 9–10 block. The reference-dependent distance measures magnitude of deviation; the anatomical capacity distributions describe the corresponding group variation (Section 3.4).

3.3 Subspace capacity varies less than individual-rank coupling

The empirical question is whether anatomical capacity differs less between exchange groups when evaluated over a subspace than at an individual rank. This hypothesis was tested for the most prevalent cluster, rank modes 9–10 (Fig. 4).

Among the 152 cluster members, 95 showed no exchange and 57 showed exchange. Median AP-inlet coupling at rank 9 decreased from 0.454 to 0.139 mm/mm (difference -0.315 ; 95% confidence interval $[-0.360, -0.262]$; $p = 2.7 \times 10^{-23}$), whereas rank-10 coupling increased from 0.163 to 0.436 mm/mm ($p = 1.7 \times 10^{-22}$). Thus, the anatomical response redistributed between neighbouring ranks.

The corresponding subspace capacity changed from 0.551 to 0.525 mm/mm (difference -0.026 ; 95% confidence interval $[-0.084, 0.015]$; rank-biserial correlation -0.21). This is approximately a 5% median decrease, compared with approximately 69% at rank 9. The Mann–Whitney test for the capacity distributions gives $p = 0.030$ (Benjamini–Hochberg-adjusted $q = 0.040$); the bootstrap interval estimates the median difference.

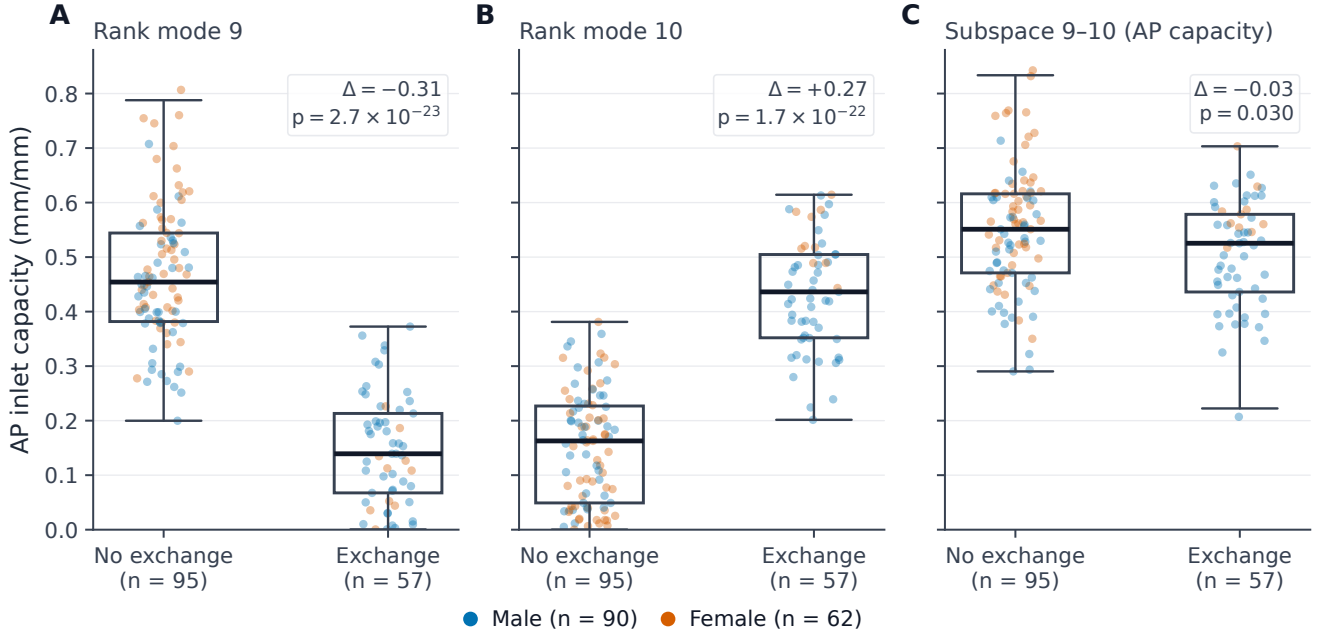


Fig. 4 Inlet coupling redistributes between rank labels in the recurrent 9–10 block. Anteroposterior (AP) inlet coupling for rank mode 9 (A), rank mode 10 (B), and the basis-invariant single-functional AP capacity of the two-dimensional 9–10 subspace (C), stratified by 9 $\rightleftharpoons$ 10 label-exchange status. Points represent individual cluster-member subjects ($n=152$; no-exchange $n=95$, exchange $n=57$; male in blue, $n=90$; female in yellow, $n=62$). Horizontal lines indicate medians; boxes indicate interquartile range; whiskers indicate $1.5 \times \text{IQR}$. Coupling is expressed as diameter change per millimetre of maximum nodal displacement.

Across all 278 subjects, irrespective of cluster membership, subspace capacity differed by -0.006 mm/mm (95% confidence interval $[-0.048, 0.019]$; $p = 0.106$), while rank-level coupling again redistributed strongly. The between-group capacity contrast is smaller than the rank-level contrast. Cluster-member estimates, effect sizes, and adjusted probabilities are provided in Table 2.

The anatomical comparison concerns capacity, whereas load routing measures actual recruitment under prescribed forces (Fig. 7). Subspace orientation and the internal 9–10 gap vary across subjects, and the block’s energy fraction under LAB_1 – LAB_3 is associated with morphology in a load-specific way. For LAB_1 , block throughput decreases with larger AP inlet diameter ($\rho_s = -0.48$, $p = 4.3 \times 10^{-17}$), whereas LAB_2 and LAB_3 show weaker and differently signed trends.

Table 2 Cluster members: rank and subspace comparisons. No exchange: $n = 95$; exchange: $n = 57$. Couplings and differences are in mm/mm . Difference is exchange minus no exchange; intervals use 5,000 percentile bootstrap resamples. p is the two-sided Mann–Whitney probability, q its Benjamini–Hochberg adjustment across the four metrics in this table, and r_{rb} the rank-biserial effect. The joint-SVD score is distinct from single-functional capacity.

Metric	No exchange	Exchange	Difference [95% CI]	r_{rb}	p	q
Rank 9	0.454	0.139	-0.315 [-0.360, -0.262]	-0.97	2.66×10^{-23}	1.07×10^{-22}
Rank 10	0.163	0.436	0.273 [0.225, 0.335]	0.95	1.70×10^{-22}	3.41×10^{-22}
$\kappa_{\text{AP}, \infty}$	0.551	0.525	-0.026 [-0.084, 0.015]	-0.21	0.030	0.040
Joint-SVD AP	0.077	0.075	-0.002 [-0.020, 0.012]	-0.02	0.840	0.840

3.4 Exploratory sex-group comparisons

Figure 5 and Table 3 describe exploratory sex-group differences in spectral and subspace measures. The sex-group tests are unadjusted exploratory comparisons.

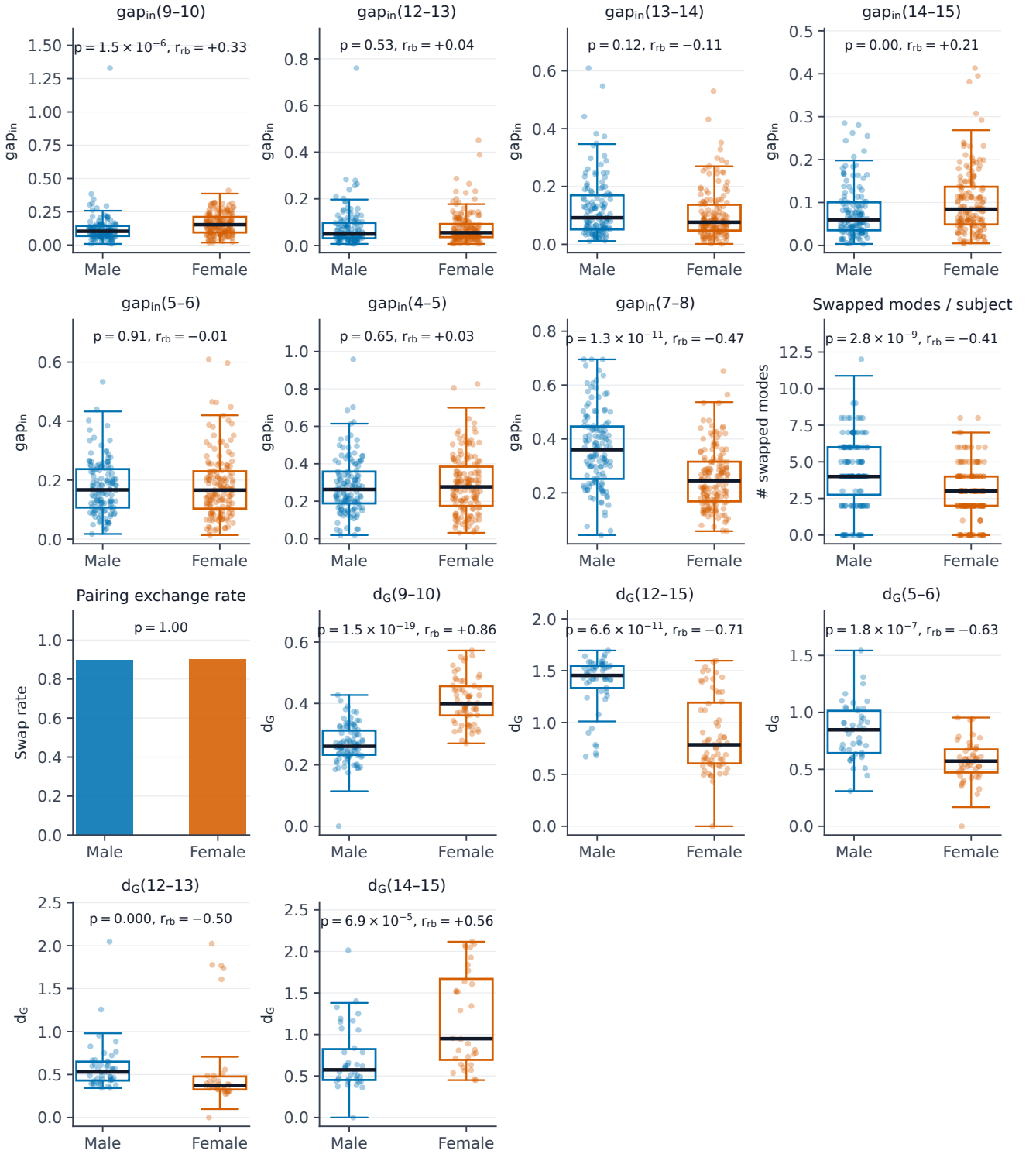


Fig. 5 Sex-stratified spectral summary

Table 3 Exploratory sex-stratified tests. Probabilities are unadjusted. Subspace distances use cluster-member subjects; other rows use the full cohort. The permutation-rate row reports a Fisher odds ratio.

Metric	Male med.	Female med.	U/OR	p	$r_{rb}/\log OR$
gap_in (9–10)	0.104	0.1534	6384	1.489e-06	0.335
gap_in (12–13)	0.04925	0.0552	9182	0.5321	0.04354
gap_in (13–14)	0.09191	0.07643	1.063e+04	0.1233	-0.1073
gap_in (14–15)	0.06002	0.08448	7631	0.003216	0.2051
gap_in (5–6)	0.1669	0.166	9677	0.9088	-0.008021
gap_in (4–5)	0.2627	0.2768	9299	0.6529	0.03135
gap_in (7–8)	0.3604	0.245	1.412e+04	1.311e-11	-0.471
swap_count_per_subject	4	3	1.352e+04	2.824e-09	-0.4079
permutation_rate	0.8984	0.9	0.9829	1	-0.01724
d_G (modes_9_10)	0.2605	0.3996	377	1.5e-19	0.8649
d_G (modes_12_15)	1.455	0.7866	2848	6.629e-11	-0.7059
d_G (modes_5_6)	0.8472	0.572	1793	1.766e-07	-0.6263
d_G (modes_12_13)	0.5299	0.3729	1120	0.0001815	-0.4973
d_G (modes_14_15)	0.5736	0.9485	270	6.922e-05	0.5559

Table 4 Dominant mode sets for the standing and proxy loading cases. N_{80} denotes the smallest set of modes required to reach 80% cumulative mean energy fraction.

Load	Top mode	Top mode share (%)	Top-3 modes	Top-3 share (%)	N_{80}	Modes in 80% set	Cumulative share at N_{80} (%)
SP2leg	2 (ML rotation)	84.49	2,4,5	93.79	1	2	84.49
SP1leg	1 (CC rotation)	39.89	1,2,3	92.04	3	1,2,3	92.04
LAB ₁	9 (inlet opening)	33.49	9,10,8	66.12	5	9,10,8,4,3	81.48
LAB ₂	5	25.15	5,6,4	74.77	4	5,6,4,9	85.8
LAB ₃	2 (ML rotation)	45.81	2,5,4	68.09	5	2,5,4,9,6	85.8

3.5 Standing and proxy loads distinguish backbone-dominated routing from higher-rank recruitment

The prescribed loads recruit different parts of the retained modal spectrum (Fig. 6, Table 4). Standing is dominated by modes 1–3. Symmetric two-leg standing (SP2leg) is concentrated in mode 2 alone, which carries about 84.5% of the retained modal strain energy; single-leg standing (SP1leg) distributes energy across the full rotational triad (modes 1–3), reaching a top-3 cumulative share of about 92.0%. Under these two prescribed standing loads, a few low-rank modes account for most of the retained modal energy.

The proxy load family recruits higher-rank modes. LAB₁, bilateral mediolateral internal ring expansion (± 400 N outward), preferentially engages the rank 9–10 inlet-related subspace, which carries a mean energy share of about 0.51 and is led by the inlet-opening mode. LAB₂, bilateral ischial-tuberosity distraction, reroutes energy into the mid-rank block at modes 4–8, where the 5–6 outlet-related pair becomes especially important. LAB₃, anteroposterior outlet distraction, is hybrid, retaining sizeable low-rank contributions while also recruiting mid-rank and 9–10 pathways. Thus, loading geometry changes the distribution of retained modal energy.

Among the recurrent clustered regimes, 9–10 remains the clearest case because it combines high cohort prevalence, clear inlet-related interpretation, direct label-versus-subspace evidence (Fig. 4), and substantial recruitment under LAB₁. The 5–6 pair provides a secondary contrast under LAB₂. The 12–15 block is common across subjects, but none of the selected loads establishes it as one dominant pathway. Primary cohort routing is evaluated on the total static response. Archived incremental outputs for 18 subjects showed unchanged top-mode identity across all five loads, but individual 9–10 block fractions shifted by up to 4.24 percentage points and dominant-set size changed in four LAB₁ cases (Section 2.3). Exploratory sex-group and capacity-load associations are reported in Section 3.4 and Appendix B.

The three prescribed traction patterns compare recruitment across pelvic loading geometries in the linear model.

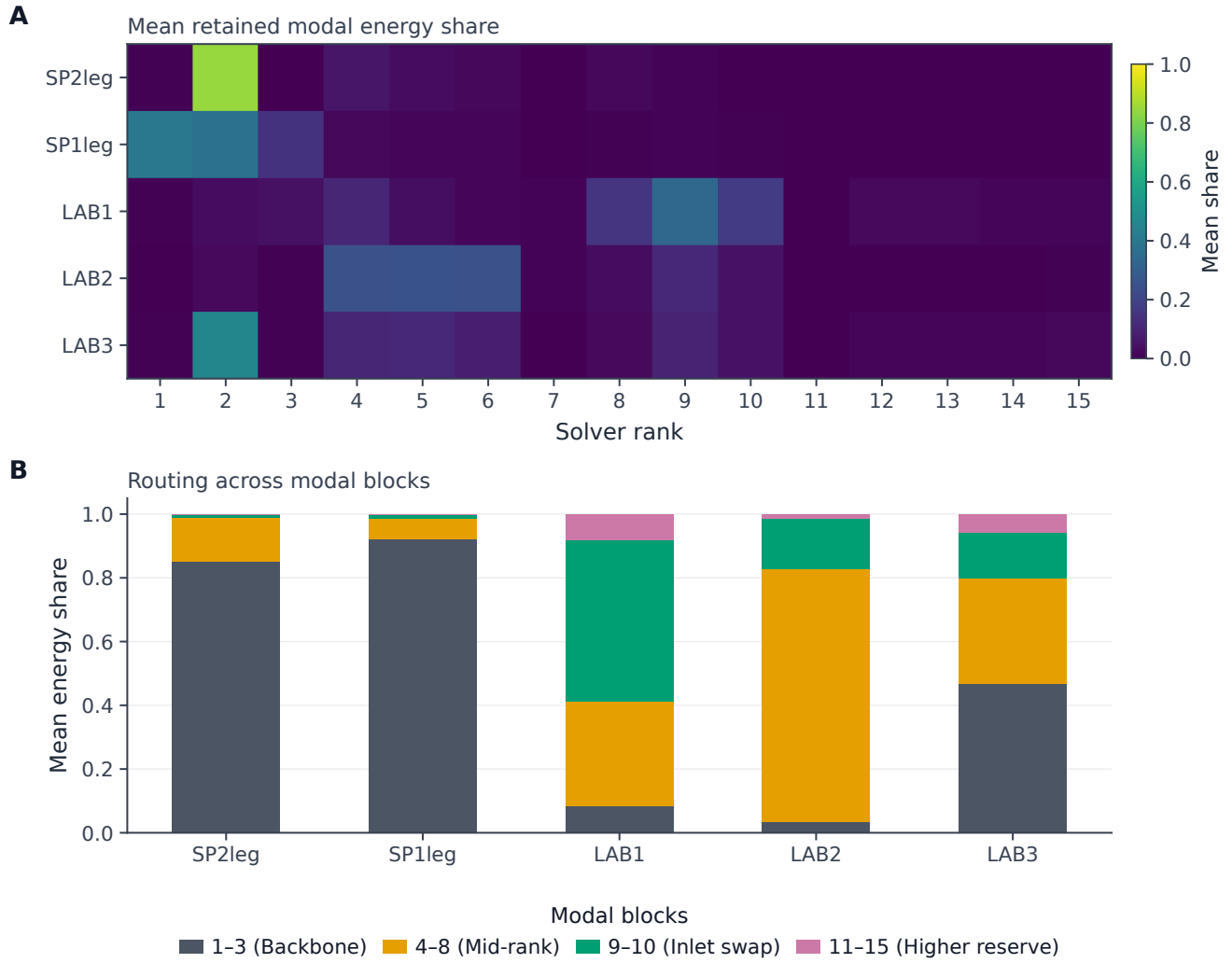


Fig. 6 Standing and comparative pelvic expansion loads recruit different modal regimes. (A) Heatmap of mean retained modal strain-energy share across the 15 lowest solver ranks for standing (SP2leg, SP1leg) and comparative pelvic expansion loads (LAB₁–LAB₃). **(B)** Partition of retained modal strain energy across functional blocks: low-rank backbone (modes 1–3; blue), mid-rank regime (modes 4–8; orange), inlet cluster (modes 9–10; green), and higher-rank reserve (modes 11–15; purple). Standing concentrates in the backbone, whereas proxy loads recruit higher ranks in a task-specific pattern.

3.6 Morphology and load routing of the 9–10 block

Figure 7 shows the association between AP inlet diameter and 9–10 block recruitment under the three comparative expansion loads, together with AP-inlet capacity by sex and label-exchange status. Capacity–load correlations across the five prevalent blocks are reported in Appendix B.

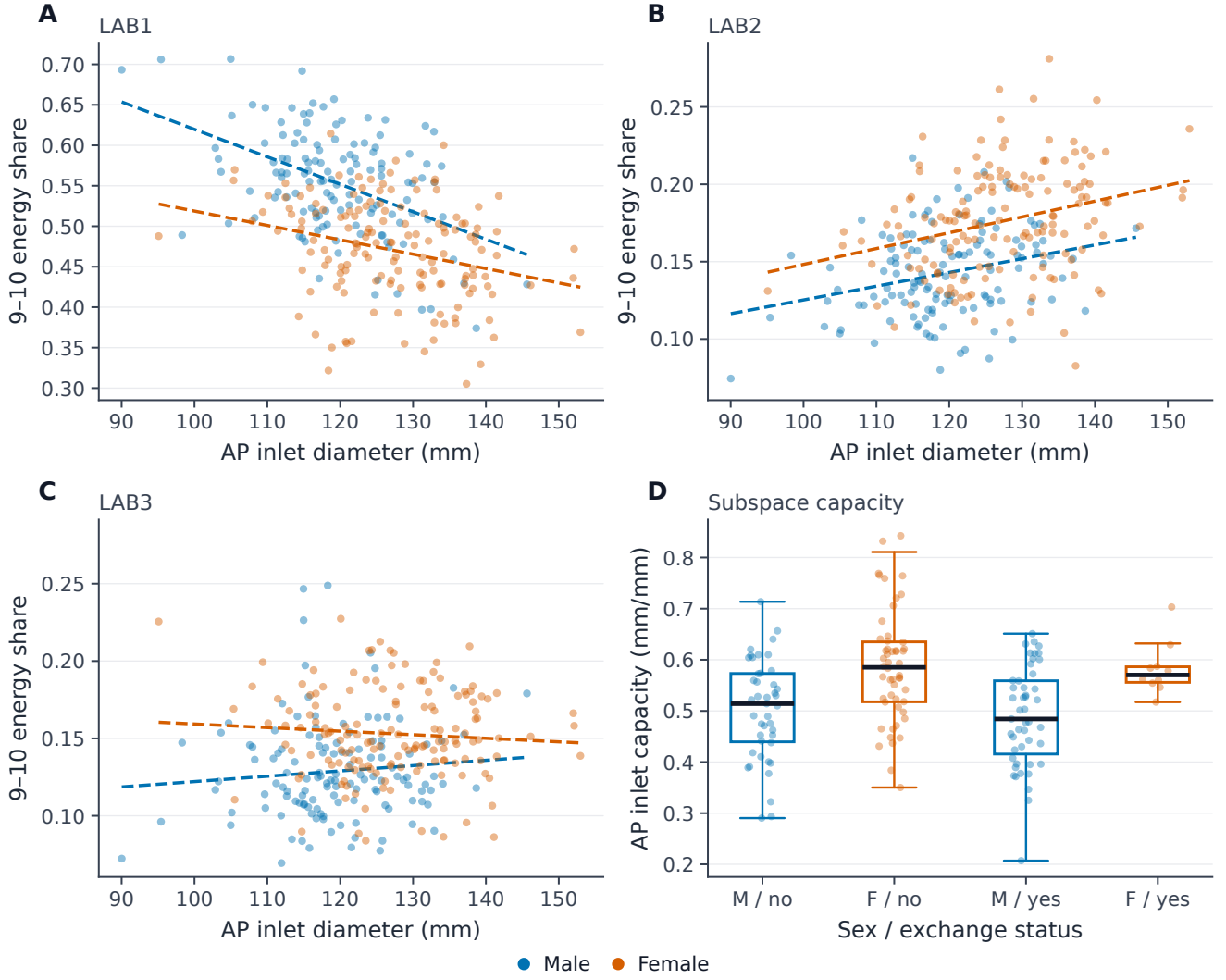


Fig. 7 Morphology and loading modulate recruitment of the 9–10 modal block. (A–C) Modal strain-energy fraction carried by the 9–10 block as a function of AP inlet diameter for LAB₁ (bilateral internal ring expansion), LAB₂ (ischial distraction), and LAB₃ (outlet distraction). Blue dots and dashed lines indicate males; yellow dots and dashed lines indicate females. (D) Basis-invariant single-functional AP-inlet capacity of the 9–10 subspace stratified jointly by sex and label-exchange status (M/no, F/no, M/yes, F/yes). Boxes show median and IQR; whiskers extend to $1.5 \times \text{IQR}$.

3.7 Model-boundary and sensitivity analyses

Figure 8 shows sensitivity to cluster threshold and reference choice, size–gap associations, and 15-mode displacement reconstruction errors (Table 5). The exploratory allometric and material-uncertainty analyses are reported in Appendices C and D.

Table 5 Relative displacement reconstruction error for 15 modes. Errors are relative displacement norms expressed as fractions.

Load	Median relative error	95th percentile	Maximum
SP2leg	0.002	0.003	0.005
SP1leg	0.002	0.003	0.004
LAB1	0.032	0.050	0.066
LAB2	0.035	0.050	0.058
LAB3	0.036	0.056	0.074

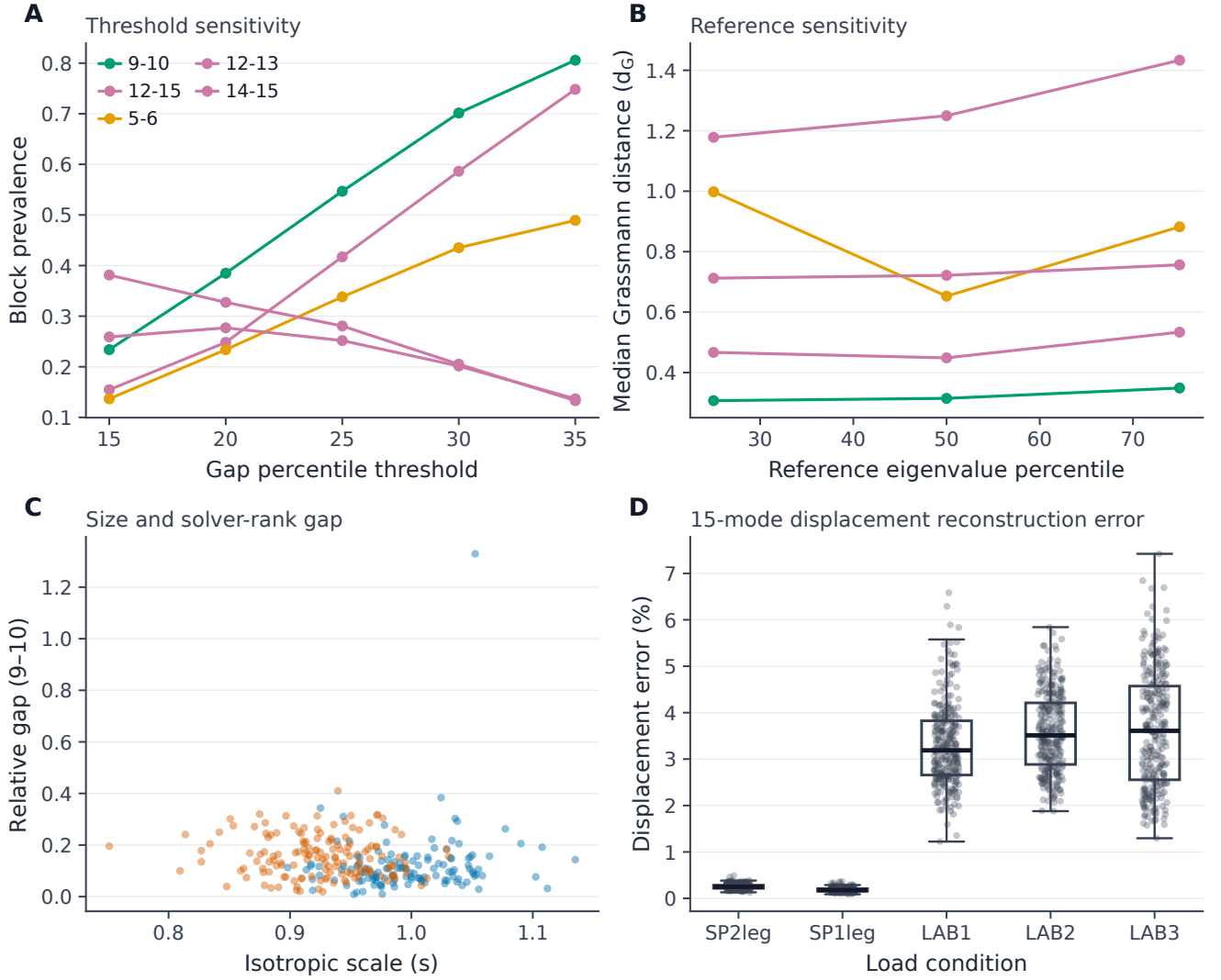


Fig. 8 Robustness and reconstruction dashboard

4 Discussion

4.1 The comparative unit depends on spectral separation

Among members of the recurrent 9–10 cluster, median AP-inlet coupling at rank 9 falls from 0.454 to 0.139 mm/mm between no-exchange and exchange groups, a difference of approximately 69%. The corresponding maximum capacity of the two-mode space falls from 0.551 to 0.525 mm/mm, approximately 5%. The capacity difference is also detectable across groups ($q = 0.040$). Together, these results show that a rank-specific comparison accentuates the anatomical contrast relative to the capacity available within the pair.

Modal assignment identifies correspondence changes, principal angles measure subspace orientation, and anatomical functionals measure diameter response. Basis invariance makes the capacity independent of the basis chosen within each subject’s subspace. Cross-subject stability is measured from the resulting capacity distributions. The exchange-group contrast is descriptive because the groups are defined using the modes whose coupling is compared.

This choice of comparison unit depends on spectral structure. Modes 1–3 rarely exchange order in the cohort, supporting individual-mode comparisons at low rank. The recurrent 9–10 pair has measurable separation on both sides and permits an interior-subspace analysis. The 12–15 block reaches the eigensolver cutoff, leaving its upper separation unknown. Gap and truncation checks therefore guide the anatomical interpretation of each block.

4.2 Anatomical capacity and load recruitment

Anatomical capacity measures the greatest modelled diameter change within a subspace per 1 mm maximum nodal displacement. Modal energy fractions measure recruitment under a prescribed load. Standing concentrates retained energy in low ranks, while internal-ring expansion recruits more of the 9–10 block. The 5–6 pair contributes to

ischial distraction despite lower outlet capacity than some other blocks. Capacity and recruitment thus provide complementary descriptions of the linear pelvic response.

These results concern initial deformation of a constrained pelvic ring and inform hypotheses about load transfer and canal change (Gruss and Schmitt 2015). The load cases use prescribed standing and expansion tractions. Their physiological interpretation is bounded by the material, joint, and boundary-condition assumptions detailed below.

4.3 Implications for population finite-element studies

Earlier pelvic work used stiffness spectra as mechanical descriptors (Henyš et al. 2021) and compared responses under specified loads (Salo et al. 2017). The present analysis connects correspondence of spectral modes to anatomically measured capacity and load recruitment across subjects. Invariant-subspace methods provide the established mathematical basis (Ghosh and Ghanem 2012); the pelvic cohort quantifies the resulting rank-versus-subspace contrast.

The same analysis sequence can be tested in other anatomical models: identify crowded eigenvalue regions, check external separation, compare subspaces in a shared metric, and evaluate anatomical observables with load recruitment. Each structure requires its own functionals and spectral-range assessment. The reference-mode atlas and exploratory sensitivity analyses are documented in the appendices. Sex-group and morphology associations are exploratory descriptions of this cohort.

4.4 Limitations and validation priorities

The FE models use small strains, isotropic bone elasticity, homogeneous cartilage, prescribed ligament properties, and a fixed superior sacral restraint. They omit subject-specific ligament laxity, pelvic floor support, nonlinear contact, and time-dependent tissue response (Vleeming et al. 2012; Chen and Grimm 2021). The mixed-sex cohort has a mean age of approximately 54 years. These assumptions define the scope of the initial-response and pelvic-loading results. The influence of the sacral penalty parameter ($\gamma = 10^6 \text{ N/mm}^3$) remains unquantified because a documented sensitivity analysis is unavailable.

Appendix D propagates assumed material covariance through first-order eigenvalue derivatives; full FE resolves would also assess eigenvectors and anatomical capacities. Appendix E illustrates possible rank redistribution under a specified stochastic model.

Modal energy fractions are normalised over the 15 retained modes. In three subjects with archived expanded spectra, these modes captured 99.3–99.5% of full quadratic energy for standing and 76.5–94.9% for expansion loads. A cohort-wide expanded-spectrum analysis would quantify truncation effects for the other subjects. Experimental ring-displacement measurements under comparable loading would test the predicted anatomical coupling and recruitment.

5 Conclusions

Across 278 pelvic FE models, modes 1–3 rarely exchange order, while anatomical coupling at higher ranks can move between neighbouring modes. In the recurrent 9–10 block, the difference in AP-inlet capacity of the two-mode space between exchange groups is much smaller than the difference assigned to rank 9. Comparing individual modes with their clustered subspaces therefore gives a fuller account of the deformation available in this linearised pelvic model.

Statements and Declarations

Funding

This research was supported by the Czech Science Foundation, Grant No. 24-10862S.

Competing interests

The authors declare no competing interests.

Ethics approval

This secondary analysis used anonymised data and precomputed models from the retrospective BoneDat cohort; no participants were recruited and no new imaging or interventions were performed. The source data collection and retrospective reuse were approved by the Ethics Committee of the University Hospital Hradec Králové (reference 202411IO3P), as reported in the BoneDat publication (Henyš and Kuchař 2025).

Consent for publication

Not applicable; no identifiable individual participant information is presented.

Data availability

The anonymised source CT scans, segmentations, and standardised geometries are publicly archived in BoneDat (<https://doi.org/10.5281/zenodo.15189761>) and described by [Henyš and Kuchař \(2025\)](#). Summary results are included in this article. The additional subject-level FE outputs and derived analyses reported here have not yet been publicly archived.

Code availability

The study uses custom Python code for model assembly, spectral analysis, and postprocessing. A public archival release of this code has not yet been established.

Author contributions

P.H., M.K., and N.H. conceptualised the study. P.H. and M.K. developed the computational models and performed the simulations. P.H., M.K., and A.M. analysed the data. P.H., M.K., A.M., and N.H. interpreted the results and contributed to manuscript writing.

A Reference-mode polynomial deformation atlas

A.1 Methods: whole-pelvis deformation decomposition

To distinguish global ring kinematics from joint-localised strains, we decompose the reference displacement fields after removing infinitesimal rigid motion. The fractions below partition the nonrigid squared displacement norm within a specified polynomial dictionary. Synthetic checks, the reference-mode atlas, and complete profile tables follow below.

A.1.1 Joint strain and global ring kinematics

The pelvic ring is a closed, doubly-curved structural biomechanical loop composed of three major bony segments (the sacrum and bilateral coxal bones) joined anteriorly at the pubic symphysis and posteriorly at the bilateral sacroiliac joints (SIJs). In continuum mechanics, the standard approach to characterise modal kinematics would be to evaluate the infinitesimal Cauchy strain tensor field,

$$\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^\top). \quad (22)$$

In the present quasi-static finite-element formulation, an effective joint modulus of $E_{\text{cartilage}} = 1.0$ MPa is prescribed for the sacroiliac cartilage cleft and pubic symphysis disc. The prescribed cartilage modulus is much lower than the representative bone moduli ($E_{\text{bone}} \sim 5\text{--}18$ GPa). In these models, strain concentrates near the compliant joints: projection of modal Cauchy strain onto low-order polynomial strain fields leaves 97–99% of the strain norm unexplained across the 15 reference modes. Such a strain-based description therefore captures little of the global ring motion of interest here.

For Mode 2, a tested one-axis dictionary left 84% of the nonrigid squared displacement norm unexplained, motivating a three-dimensional basis that represents coupled flexure and shear. We therefore analyse displacement over the full three-dimensional pelvic domain. After removing infinitesimal rigid motion, we project the remaining field onto four polynomial deformation families. Their unexplained residual remains explicit.

A.1.2 Continuum inner product and tetrahedral quadrature

Let $\Omega \subset \mathbb{R}^3$ denote the reference pelvic continuum domain with volume $V = \int_{\Omega} dV$. The natural inner product on the displacement space $[L^2(\Omega)]^3$ weighted by volume is

$$\begin{aligned} \langle \mathbf{u}, \mathbf{v} \rangle_V &= \frac{1}{V} \int_{\Omega} \mathbf{u}(\mathbf{x}) \cdot \mathbf{v}(\mathbf{x}) dV, \\ \|\mathbf{u}\|_V^2 &= \langle \mathbf{u}, \mathbf{u} \rangle_V. \end{aligned} \quad (23)$$

The continuous reference domain Ω is discretized into an unstructured tetrahedral mesh $\mathcal{T}_h = \{K_e\}_{e=1}^{N_e}$ using piecewise-linear (P_1) Lagrange continuous finite elements. Within any tetrahedron K , a P_1 displacement field

Volume integration of modal and polynomial fields

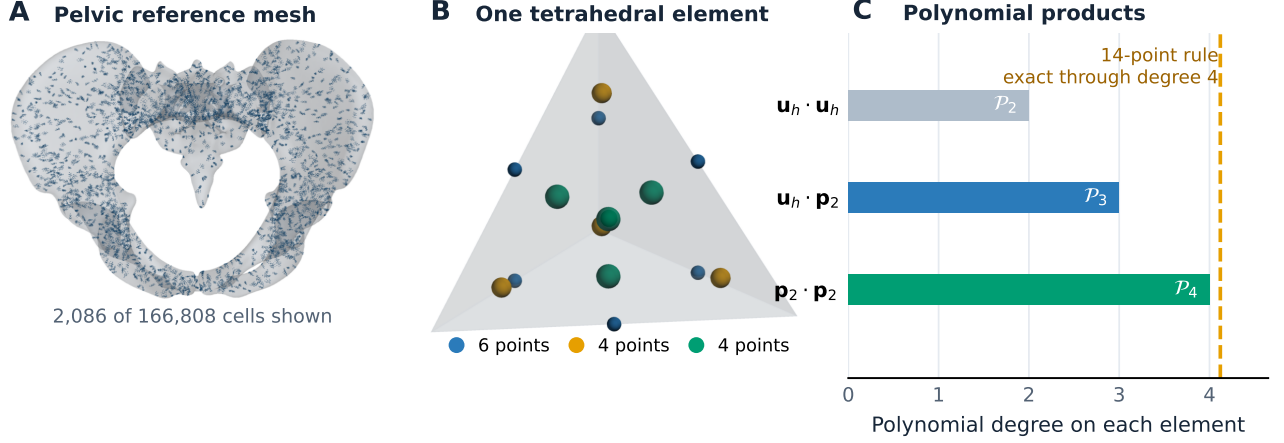


Fig. 9 Volume quadrature for the whole-pelvis displacement decomposition. (A) Reference pelvic mesh with quadrature points displayed for every 80th tetrahedron; the calculation includes all 166 808 cells. (B) The 14 integration points on a standard tetrahedron. Colour and sphere size distinguish the three groups of positive weights (six, four, and four points). The rule is mapped to every tetrahedron in the mesh. (C) Polynomial degrees of the squared P_1 FE displacement, its cross-product with a quadratic dictionary field, and the quadratic dictionary Gram product. The 14-point rule is exact through degree 4 on each tetrahedron.

$\mathbf{u}_h(\mathbf{x})$ is an affine vector function of position ($\mathbf{u}_h|_K \in \mathcal{P}_1(K)$). When computing projections against quadratic deformation modes $\mathbf{p}_2 \in \mathcal{P}_2(K)$, the modal cross-product integrand $\mathbf{u}_h \cdot \mathbf{p}_2$ is a cubic polynomial ($\mathcal{P}_3(K)$), and the dictionary Gram matrix integrand $\mathbf{p}_2 \cdot \mathbf{p}_2$ is a quartic polynomial ($\mathcal{P}_4(K)$).

The four-point tetrahedral rule integrates squared P_1 displacement exactly ($\mathcal{P}_2(K)$). Products of quadratic dictionary fields require degree-4 exactness. For the reference atlas and its accompanying table, we used a positive 14-point tetrahedral quadrature rule exact through polynomial degree 4. For a rule exact to polynomial degree d ($d = 2$ for four points, $d = 4$ for 14 points), the quadrature points $\mathbf{x}_{K,q}$ and strictly positive weights $w_{K,q} > 0$ satisfy $\sum_{q=1}^{N_{q,\text{tet}}} w_{K,q} = |K|$:

$$\int_K p(\mathbf{x}) dV = \sum_{q=1}^{N_{q,\text{tet}}} w_{K,q} p(\mathbf{x}_{K,q}) \quad \forall p \in \mathcal{P}_d(K). \quad (24)$$

Summing across all N_e tetrahedra gives $N_q = N_{q,\text{tet}} N_e$ discrete sampling points with weights $w_p > 0$ ($N_q = 4N_e$ for the 4-point rule; $N_q = 14N_e$ for the degree-4 rule). Defining normalised quadrature weights $\tilde{w}_p = w_p / \sum_{j=1}^{N_q} w_j$, the continuum inner product (23) is evaluated as:

$$\langle \mathbf{u}, \mathbf{v} \rangle_V \approx \sum_{p=1}^{N_q} \tilde{w}_p \mathbf{u}(\mathbf{x}_p) \cdot \mathbf{v}(\mathbf{x}_p). \quad (25)$$

The degree-4 rule integrates the products of quadratic dictionary functions exactly on each tetrahedron. Because the reference modes are P_1 , it also integrates their cross-products with the dictionary exactly. The degree-4 rule integrates the quadratic-dictionary projection exactly on each tetrahedron and supplies positive weights for the inner product. A four-point, degree-2 sensitivity recomputation of all 15 reference modes changed the deformation-family and residual fractions by at most 1.91×10^{-5} percentage points. Figure 9 links the element rule to the whole-domain inner product and to the degrees of the three products used in the decomposition.

A.1.3 Infinitesimal rigid-body motion elimination

Infinitesimal rigid-body displacements in $\mathbb{R}^3$ constitute a 6-dimensional linear Lie algebra subspace $\mathfrak{se}(3)$ consisting of 3 translations and 3 rotations:

$$\begin{aligned} \mathbf{u}_{\text{rigid}}(\mathbf{x}) &= \mathbf{t} + \boldsymbol{\omega} \times (\mathbf{x} - \mathbf{x}_c), \\ \mathbf{t}, \boldsymbol{\omega} &\in \mathbb{R}^3, \end{aligned} \quad (26)$$

where $\mathbf{x}_c$ is the volume-weighted spatial centroid of the pelvic domain,

$$\mathbf{x}_c = \frac{1}{V} \int_{\Omega} \mathbf{x} dV = \sum_{p=1}^{N_q} \tilde{w}_p \mathbf{x}_p. \quad (27)$$

To ensure scale-free, dimensionally consistent kinematics across subjects, we define the reference gyration radius r_g and non-dimensionalised anatomical coordinates $\tilde{\mathbf{x}} = (\tilde{x}, \tilde{y}, \tilde{z})^\top$:

$$r_g = \sqrt{\frac{1}{V} \int_{\Omega} \|\mathbf{x} - \mathbf{x}_c\|^2 dV} = \sqrt{\sum_{p=1}^{N_q} \tilde{w}_p \|\mathbf{x}_p - \mathbf{x}_c\|^2}, \quad (28)$$

$$\tilde{\mathbf{x}} = \frac{\mathbf{x} - \mathbf{x}_c}{r_g}.$$

Here the Cartesian coordinate directions correspond to the standard anatomical axes: x is mediolateral (ML), y is anteroposterior (AP), and z is craniocaudal (CC).

In the discrete quadrature representation, each displacement field is a stacked vector $\mathbf{U} \in \mathbb{R}^{3N_q}$ with components $U_{3p-3+i} = \sqrt{\tilde{w}_p} u_i(\mathbf{x}_p)$ for $i \in \{1, 2, 3\}$. The 6 rigid-body design modes $\mathbf{r}_1, \dots, \mathbf{r}_6 \in \mathbb{R}^{3N_q}$ (corresponding to unit translations along x, y, z and unit infinitesimal rotations about x, y, z) form the rigid design matrix $\mathbf{R} \in \mathbb{R}^{3N_q \times 6}$. We compute its thin QR decomposition,

$$\mathbf{R} = \mathbf{Q}_{\text{rigid}} \mathbf{R}_{\text{upper}}, \quad (29)$$

$$\mathbf{Q}_{\text{rigid}} \in \mathbb{R}^{3N_q \times 6}, \quad \mathbf{Q}_{\text{rigid}}^\top \mathbf{Q}_{\text{rigid}} = \mathbf{I}_6.$$

The orthogonal projection operator onto the nonrigid displacement subspace is then:

$$\mathbf{P}_{\text{nonrigid}} = \mathbf{I}_{3N_q} - \mathbf{Q}_{\text{rigid}} \mathbf{Q}_{\text{rigid}}^\top. \quad (30)$$

For a sampled modal field $\mathbf{U}$, define $\mathbf{U}_{\text{nr}} = \mathbf{P}_{\text{nonrigid}} \mathbf{U}$ and $\mathbf{U}_{\text{rigid}} = \mathbf{U} - \mathbf{U}_{\text{nr}}$. Because the quadrature weights are incorporated into $\mathbf{U}$, its Euclidean norm equals the quadrature-evaluated continuum norm. Orthogonality gives

$$\|\mathbf{U}\|_2^2 = \|\mathbf{U}_{\text{rigid}}\|_2^2 + \|\mathbf{U}_{\text{nr}}\|_2^2, \quad \|\mathbf{U}\|_2^2 = \|\mathbf{u}\|_V^2, \quad (31)$$

$$f_{\text{nonrigid}} = \frac{\|\mathbf{U}_{\text{nr}}\|_2^2}{\|\mathbf{U}\|_2^2}.$$

The decomposition acts on $\mathbf{U}_{\text{nr}}$ and excludes the projected rigid component. A field with $f_{\text{nonrigid}} \leq 10^{-20}$ is classified as rigid and excluded from the nonrigid decomposition.

A.1.4 Canonical 3D polynomial dictionary

The vector space of polynomial displacement fields $\mathbf{u}(\tilde{\mathbf{x}})$ of degree at most 2 in three dimensions has complete algebraic dimension $3 \times \binom{3+2}{2} = 30$. Subtracting the 6 degrees of freedom of infinitesimal rigid motion leaves an exact 24-dimensional nonrigid polynomial quotient space. We partition this polynomial quotient into four kinematic families with a total of 24 independent degrees of freedom:

- (i) **Axial family** ($d_{\text{axial}} = 6$ degrees of freedom): Uniform normal extension/compression along each anatomical axis, together with quadratic normal strain gradients:

$$\begin{aligned} \mathbf{a}_{x,1} &= (\tilde{x}, 0, 0)^\top, \\ \mathbf{a}_{x,2} &= (\tfrac{1}{2}\tilde{x}^2, 0, 0)^\top, \\ \mathbf{a}_{y,1} &= (0, \tilde{y}, 0)^\top, \\ \mathbf{a}_{y,2} &= (0, \tfrac{1}{2}\tilde{y}^2, 0)^\top, \\ \mathbf{a}_{z,1} &= (0, 0, \tilde{z})^\top, \\ \mathbf{a}_{z,2} &= (0, 0, \tfrac{1}{2}\tilde{z}^2)^\top. \end{aligned} \quad (32)$$

- (ii) **Bending family** ($d_{\text{bending}} = 6$ degrees of freedom): Shear-free quadratic flexural fields of Euler–Bernoulli form. For any pair of distinct coordinate axes $i \neq j \in \{x, y, z\}$, the displacement field

$$\mathbf{b}_{ij} = \tfrac{1}{2}\tilde{x}_i^2 \mathbf{e}_j - \tilde{x}_i \tilde{x}_j \mathbf{e}_i \quad (33)$$

satisfies $\partial_i u_j = \tilde{x}_i$ and $\partial_j u_i = -\tilde{x}_i$. Consequently, the Cauchy shear strain vanishes identically:

$$\varepsilon_{ij}(\mathbf{b}_{ij}) = \tfrac{1}{2}(\partial_j u_i + \partial_i u_j) = \tfrac{1}{2}(-\tilde{x}_i + \tilde{x}_i) \equiv 0. \quad (34)$$

The six ordered pairs (x, y) , (x, z) , (y, x) , (y, z) , (z, x) , (z, y) thus define six independent shear-free flexural modes.

- (iii) **Shear family** ($d_{\text{shear}} = 10$ **degrees of freedom**): Comprises three uniform transverse shears, six planar quadratic transverse shear gradients, and one triaxial cross-gradient shear field $\mathbf{w}$:

$$\begin{aligned}
\mathbf{s}_{xy,1} &= (\tilde{y}, \tilde{x}, 0)^\top, \\
\mathbf{s}_{xy,2} &= (\tilde{x}\tilde{y}, \frac{1}{2}\tilde{x}^2, 0)^\top, \\
\mathbf{s}_{yx,2} &= (\frac{1}{2}\tilde{y}^2, \tilde{y}\tilde{x}, 0)^\top, \\
\mathbf{s}_{yz,1} &= (0, \tilde{z}, \tilde{y})^\top, \\
\mathbf{s}_{yz,2} &= (0, \tilde{y}\tilde{z}, \frac{1}{2}\tilde{y}^2)^\top, \\
\mathbf{s}_{zy,2} &= (0, \frac{1}{2}\tilde{z}^2, \tilde{z}\tilde{y})^\top, \\
\mathbf{s}_{zx,1} &= (\tilde{z}, 0, \tilde{x})^\top, \\
\mathbf{s}_{zx,2} &= (\frac{1}{2}\tilde{z}^2, 0, \tilde{z}\tilde{x})^\top, \\
\mathbf{s}_{xz,2} &= (\tilde{x}\tilde{z}, 0, \frac{1}{2}\tilde{x}^2)^\top, \\
\mathbf{w} &= (\tilde{y}\tilde{z}, \tilde{x}\tilde{z}, \tilde{x}\tilde{y})^\top = \nabla(\tilde{x}\tilde{y}\tilde{z}).
\end{aligned} \tag{35}$$

Unlike bending, these modes produce non-zero shear strains $\varepsilon_{ij} \neq 0$. The triaxial mode $\mathbf{w}$ is irrotational ($\nabla \times \mathbf{w} = \mathbf{0}$) and divergence-free ($\nabla \cdot \mathbf{w} = 0$), completing the chosen basis of the 24-dimensional nonrigid polynomial quotient.

- (iv) **Torsion family** ($d_{\text{torsion}} = 2$ **degrees of freedom**): Polynomial twist-gradient fields around the three Cartesian axes:

$$\mathbf{t}_x = (0, -\tilde{x}\tilde{z}, \tilde{x}\tilde{y})^\top, \quad \mathbf{t}_y = (\tilde{y}\tilde{z}, 0, -\tilde{x}\tilde{y})^\top, \quad \mathbf{t}_z = (-\tilde{y}\tilde{z}, \tilde{x}\tilde{z}, 0)^\top. \tag{36}$$

Each torsion field exhibits identically zero normal strain ($\varepsilon_{xx} = \varepsilon_{yy} = \varepsilon_{zz} \equiv 0$). Crucially, these three vector fields satisfy the identity:

$$\mathbf{t}_x + \mathbf{t}_y + \mathbf{t}_z \equiv \mathbf{0}. \tag{37}$$

Hence, this selected torsion-like polynomial family has algebraic dimension 2, spanned by $\{\mathbf{t}_x, \mathbf{t}_y\}$.

Figure 10 shows one rigid-projected polynomial field from each family on the reference anatomy.

To assemble the discrete kinematic design matrix, the basis fields in each family $m \in \{\text{axial, bending, shear, torsion}\}$ are sampled at the N_q quadrature points and weighted by $\sqrt{\bar{w}_p}$ to form $\mathbf{A}_m^{\text{raw}}$. They are projected orthogonal to $\mathbf{Q}_{\text{rigid}}$ and orthonormalised via reduced singular value decomposition (SVD):

$$\begin{aligned}
\mathbf{P}_{\text{nonrigid}} \mathbf{A}_m^{\text{raw}} &= \mathbf{Q}_m \Sigma_m \mathbf{V}_m^\top, \\
\mathbf{Q}_m &\in \mathbb{R}^{3N_q \times d_m}, \quad \mathbf{Q}_m^\top \mathbf{Q}_m = \mathbf{I}_{d_m}.
\end{aligned} \tag{38}$$

The concatenated design matrix $\mathbf{A} = [\mathbf{Q}_{\text{axial}}, \mathbf{Q}_{\text{bending}}, \mathbf{Q}_{\text{shear}}, \mathbf{Q}_{\text{torsion}}] \in \mathbb{R}^{3N_q \times 24}$ spans the complete 24-dimensional canonical nonrigid deformation subspace.

A.1.5 Numerical conditioning, family non-orthogonality, and the Gram matrix

The four family subspaces are not mutually orthogonal even on an idealised symmetric domain. For example, on the symmetric unit cube $[-1, 1]^3$ (volume $V = 8$), the unnormalised continuum volume integral between the bending mode $\mathbf{b} = (-xy, \frac{1}{2}x^2, 0)^\top$ and the shear mode $\mathbf{s} = (xy, \frac{1}{2}x^2, 0)^\top$ evaluates analytically to $\int_{[-1,1]^3} \mathbf{b} \cdot \mathbf{s} dV = -22/45 \approx -0.4889$, corresponding to a volume-normalised inner product $\langle \mathbf{b}, \mathbf{s} \rangle_V = \frac{1}{V} \int_{[-1,1]^3} \mathbf{b} \cdot \mathbf{s} dV = -11/180 \approx -0.0611$ (angular cosine $\cos \theta = -11/29 \approx -0.3793$). Both example fields have the same nonzero mean translation $(0, 1/6, 0)^\top$. After subtracting it, their volume-normalised inner product is $-4/45$ and their cosine is $-2/3$. Thus non-orthogonality also holds after the rigid-motion projection used here.

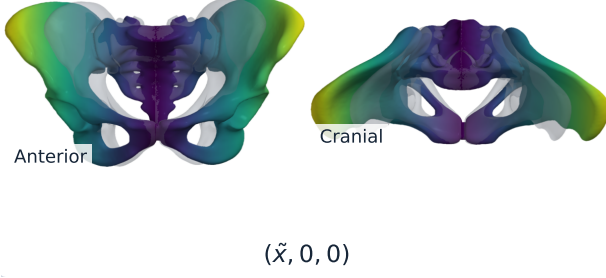
On the real, geometrically asymmetric pelvic ring, this intrinsic non-orthogonality combines with anatomical geometry. The kinematic design matrix $\mathbf{A}$ has 2-norm condition number $\kappa(\mathbf{A}) \approx 4.58$, and the associated Gram matrix $\mathbf{G} = \mathbf{A}^\top \mathbf{A} \in \mathbb{R}^{24 \times 24}$ has condition number:

$$\kappa(\mathbf{G}) = \frac{\sigma_{\max}(\mathbf{G})}{\sigma_{\min}(\mathbf{G})} = \kappa(\mathbf{A})^2 \approx 20.98. \tag{39}$$

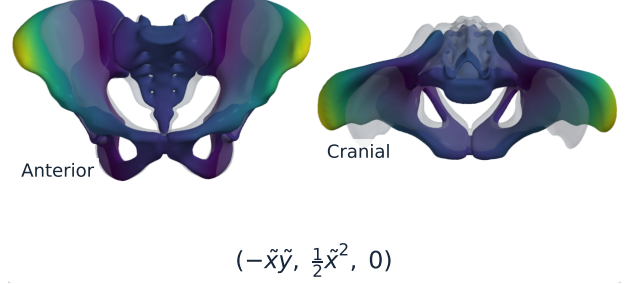
At the reference geometry, the sampled design columns are numerically independent (Gram condition number ~ 21). Family shares remain conditional on the selected dictionary.

Representative fields in the 3D polynomial dictionary

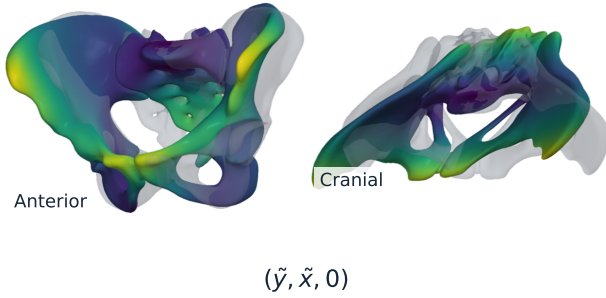
A Axial · Normal extension



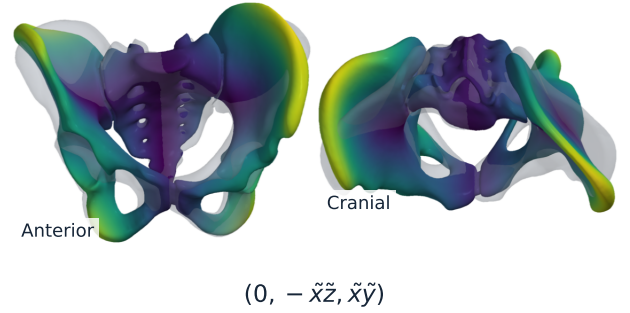
B Bending · Shear-free flexure



C Shear · Transverse shear



D Torsion · Twist gradient



Ghost: undeformed reference · Colour: normalized displacement magnitude · Each field scaled to 50 mm for display

Fig. 10 Representative fields of the 3D polynomial deformation dictionary. One example per family is evaluated on the reference pelvis: (A) axial extension, $\mathbf{a}_{x,1}$; (B) shear-free bending, $\mathbf{b}_{xy}$; (C) uniform transverse shear, $\mathbf{s}_{xy,1}$; and (D) torsion-like twist, $\mathbf{t}_x$. Each card shows anterior and cranial views. Grey denotes the undeformed geometry; colour indicates displacement magnitude relative to the maximum within that field. The formulas are the raw polynomial dictionary members in dimensionless coordinates, and the rendered fields have their volume-weighted infinitesimal rigid component removed. Each representative basis field is independently scaled to a 50 mm maximum displacement for visibility.

A.1.6 Cooperative game-theoretic Shapley value norm allocation

Because the four family subspaces are not mutually orthogonal, sequential projection would allocate correlated fitted norm according to family order. We use Shapley allocation to average marginal contributions over all orders. To obtain an order-independent allocation conditional on this chosen dictionary, we formulate the problem using cooperative game theory.

Let $M = \{\text{axial}, \text{bending}, \text{shear}, \text{torsion}\}$ denote the set of four pattern players ($|M| = 4$). There are $2^4 = 16$ possible coalitions $S \subseteq M$. For any coalition S , $\mathbf{A}_S$ concatenates the orthonormal family matrices $\mathbf{Q}_m$ for $m \in S$ and has $\sum_{m \in S} d_m$ columns. We define the characteristic function $v(S)$ as the fraction of nonrigid squared displacement norm explained by their joint span:

$$v(S) = R^2(S) = \frac{\mathbf{U}_{\text{nr}}^\top \mathbf{P}_S \mathbf{U}_{\text{nr}}}{\|\mathbf{U}_{\text{nr}}\|_2^2}, \quad (40)$$

$$\mathbf{P}_S = \mathbf{A}_S (\mathbf{A}_S^\top \mathbf{A}_S)^+ \mathbf{A}_S^\top,$$

with $v(\emptyset) = 0$ and $V_S = \text{range}(\mathbf{A}_S)$; $(\cdot)^+$ denotes the Moore–Penrose pseudoinverse. The value is defined only when $\|\mathbf{U}_{\text{nr}}\|_2 > 0$. For a given coalition value function v , the allocation $f_m(v)$ satisfies the following axioms:

1. **Efficiency:** The sum of allocated values equals the value of the grand coalition:

$$\sum_{m \in M} f_m = v(M) = R^2(M) = 1 - f_{\text{residual}}. \quad (41)$$

2. **Symmetry:** If two families $m, k \in M$ contribute identically to every coalition ($v(S \cup \{m\}) = v(S \cup \{k\})$ for all $S \subseteq M \setminus \{m, k\}$), then $f_m = f_k$.

3. **Dummy player:** If a family m adds zero marginal variance to every coalition ($v(S \cup \{m\}) = v(S)$ for all $S \subseteq M \setminus \{m\}$), then $f_m = 0$.
4. **Additivity:** For any two characteristic value functions v and w defined on the same set of players, the values allocated under the combined sum game $(v + w)(S) = v(S) + w(S)$ satisfy $f_m(v + w) = f_m(v) + f_m(w)$.

The explicit Shapley allocation formula evaluates the marginal contribution of family m averaged over all coalition orders:

$$f_m = \sum_{S \subseteq M \setminus \{m\}} \frac{|S|!(|M| - |S| - 1)!}{|M|!} [v(S \cup \{m\}) - v(S)]. \quad (42)$$

The unexplained shape change is retained as an explicit residual:

$$f_{\text{residual}} = 1 - v(M) = 1 - R^2(\text{axial} \cup \text{bending} \cup \text{shear} \cup \text{torsion}). \quad (43)$$

Because subspace projection operators are monotonic ($V_S \subseteq V_{S \cup \{m\}} \implies v(S \cup \{m\}) \geq v(S)$), every marginal contribution is non-negative, which guarantees $f_m \geq 0$ for all m . Exact closure holds identically:

$$f_{\text{axial}} + f_{\text{bending}} + f_{\text{shear}} + f_{\text{torsion}} + f_{\text{residual}} \equiv 1 \quad (100\%). \quad (44)$$

Furthermore, the fractions are strictly scale-invariant ($\mathbf{u} \mapsto c\mathbf{u}$ for $c \neq 0$) and sign-invariant ($\mathbf{u} \mapsto -\mathbf{u}$). For comparative diagnostics, we also compute the standalone explained-norm fraction for each isolated family:

$$R_m^2 = v(\{m\}) = \frac{\mathbf{U}_{\text{nr}}^\top \mathbf{P}_m \mathbf{U}_{\text{nr}}}{\|\mathbf{U}_{\text{nr}}\|_2^2}. \quad (45)$$

Because the families are not mutually orthogonal, $\sum_{m \in M} R_m^2$ can exceed the jointly explained fraction $R^2(M)$. The Shapley values allocate shared explanatory content in an order-independent way conditional on the selected dictionary and anatomical axes.

A.2 Results: reference deformation atlas and numerical validation

We first check the polynomial decomposition on synthetic fields, then use it to describe the 15 reference modes. Tables 6–7 give the complete reference profiles and synthetic checks on a cube and the reference pelvic mesh; the formulation is given in Methods.

A.2.1 Synthetic checks of the deformation decomposition

Synthetic polynomial fields and rigid motions on a cube and the reference pelvic mesh provide implementation checks (Table 7).

On the symmetric domain, pure synthetic fields belonging to the axial, shear, and torsion families are identified with 100.0% fidelity, zero cross-talk, and exactly 0.0% residual. For pure Euler-Bernoulli bending ($\mathbf{u} = (-xy, \frac{1}{2}x^2, 0)$), the displacement field consists of an axial cross-term $-xy$ and a transverse parabolic deflection $\frac{1}{2}x^2$. Because the quadratic shear mode $\mathbf{s}_{xy,2} = (xy, \frac{1}{2}x^2, 0)$ shares the identical transverse parabolic deflection $\frac{1}{2}x^2$, the pure flexural field lies in the joint span $V_{\text{bending}} \oplus V_{\text{shear}}$. The equal-weight $15 \times 15 \times 15$ grid benchmark allocates 77.5% to bending and 22.5% to shear, with residual rounding to zero. The percentages follow from the sampled grid and the specified Shapley allocation.

On the actual reference pelvis mesh, pure canonical polynomial fields are recovered with exactly 0.0% residual (100.0% fitted fraction):

- Pure axial stretch ($x\mathbf{e}_x$): 85.3% axial share, 0.0% residual (100.0% fitted).
- Pure triaxial extension ($x\mathbf{e}_x - \frac{1}{2}y\mathbf{e}_y - \frac{1}{2}z\mathbf{e}_z$): 84.8% axial share, 0.0% residual (100.0% fitted).
- Pure Euler-Bernoulli bending: 74.1% bending share, 0.0% residual (100.0% fitted).
- Pure transverse shear: 82.8% shear share, 0.0% residual (100.0% fitted).
- Pure torsion-like polynomial field: 87.1% torsion share, 0.0% residual (100.0% fitted).

For pure infinitesimal translations and rotations, the computed nonrigid fraction is $f_{\text{nonrigid}} < 10^{-27}$. Finally, adding arbitrary rigid-body translations and rotations to an axial deformation field yields identical Shapley shares (85.3% axial, 7.2% bending, 7.5% shear, 0.0% torsion, 0.0% residual) to within machine precision ($< 10^{-10}$), confirming invariance to added infinitesimal rigid displacement fields.

A.2.2 Reference mode deformation profiles and biomechanical interpretation

Table 6 compiles the quantitative whole-pelvis 3D deformation decomposition for all fifteen reference eigenmodes. The reference profiles show how the chosen polynomial families partition each mode’s nonrigid squared displacement norm:

- **Modes 1 and 2 (low-rank backbone):** For Mode 1, the nonrigid component accounts for 3.3% of the total squared displacement norm; the rigid-motion projection accounts for the remainder. Its nonrigid shape change is predominantly out-of-plane torsional twist (22.7%) and transverse shear (17.7%), with a total fitted fraction of 42.7% (residual 57.3%). For Mode 2, the nonrigid component (5.5% of total squared displacement norm) is a coupled sagittal ring flexure and shear (23.6% bending, 23.7% shear, 20.8% axial, 68.6% fitted, residual 31.4%).
- **Modes 4 and 5 (dominant flexural pathways):** Modes 4 and 5 have bending shares of 43.6% and 47.4% of their nonrigid squared displacement norms, respectively (with standalone bending fits R^2_{bending} reaching 47.6% and 62.4%, and total explained fractions of 74.4% and 86.3%). These modes capture lateral and anteroposterior flaring of the iliac wings.
- **The inlet-related 9–10 pair:** Mode 9 has the highest fitted fraction in the reference spectrum (91.8% explained; 8.2% residual). Its deformation is overwhelmingly axial (60.6% Shapley share; standalone $R^2_{\text{axial}} = 67.7\%$), consistent with Cartesian normal-extension content. Its AP-inlet response is quantified separately by the anatomical diameter functional. In contrast, its near-degenerate partner Mode 10 (74.9% fitted) is dominated by shear (35.2%) and torsion (34.3%), with minimal axial contribution (4.3%; standalone $R^2_{\text{torsion}} = 30.3\%$, $R^2_{\text{shear}} = 32.7\%$, residual 25.1%). Cohort-level comparison of the 9–10 pair is reported in Section 3.3.

Table 6 Reference mode whole-pelvis 3D deformation decomposition.

Mode	Nonrigid	Fitted	Shapley norm partition (%)					Standalone R^2 (%)			
			Axial	Bending	Shear	Torsion	Residual	Axial	Bending	Shear	Torsion
1	3.3	42.7	0.9	1.5	17.7	22.7	57.3	0.4	2.2	15.4	19.0
2	5.5	68.6	20.8	23.6	23.7	0.5	31.4	15.4	16.1	16.9	0.8
3	29.0	76.7	2.8	4.7	41.7	27.5	23.3	5.1	5.1	52.7	37.4
4	24.5	74.4	13.5	43.6	15.8	1.5	25.6	24.6	47.6	14.9	2.2
5	73.0	86.3	4.3	47.4	24.2	10.4	13.7	5.4	62.4	38.7	13.7
6	71.9	82.3	3.1	26.9	24.9	27.3	17.7	2.2	33.0	37.2	35.2
7	69.5	72.1	16.9	7.3	44.3	3.5	27.9	29.4	4.6	55.8	4.1
8	57.0	72.1	22.1	20.3	29.5	0.2	27.9	32.7	32.5	41.3	0.4
9	92.1	91.8	60.6	18.4	12.2	0.5	8.2	67.7	31.6	14.9	0.3
10	95.7	74.9	4.3	1.1	35.2	34.3	25.1	6.2	1.5	32.7	30.3
11	71.4	43.1	5.4	3.9	21.1	12.8	56.9	2.8	0.4	9.5	12.2
12	94.2	62.3	9.6	22.4	27.4	3.0	37.7	13.1	12.7	20.3	2.4
13	98.3	48.4	9.2	19.9	19.1	0.2	51.6	8.3	27.0	25.7	0.0
14	92.0	53.4	1.2	15.9	33.4	3.0	46.6	0.9	5.6	24.9	4.7
15	98.7	55.6	18.4	23.5	13.7	0.0	44.4	16.1	28.2	13.7	0.0

Table 7 Synthetic checks of the deformation decomposition.

Domain	Synthetic deformation field	Axial (%)	Bending (%)	Shear (%)	Torsion (%)	Resid. (%)	Outcome
Symmetric cube	Pure axial stretch ($x\mathbf{e}_x$)	100.0	0.0	0.0	0.0	0.0	Exact closure (0.0% residual)
Symmetric cube	Pure bending ($-xy\mathbf{e}_x + \frac{1}{2}x^2\mathbf{e}_y$)	0.0	77.5	22.5	0.0	0.0	Exact closure (0.0% residual)
Symmetric cube	Pure shear ($y\mathbf{e}_x + x\mathbf{e}_y$)	0.0	0.0	100.0	0.0	0.0	Exact closure (0.0% residual)
Symmetric cube	Pure torsion ($yz\mathbf{e}_x - xy\mathbf{e}_z$)	0.0	0.0	0.0	100.0	0.0	Exact closure (0.0% residual)
Pelvis mesh	Pure axial stretch ($x\mathbf{e}_x$)	85.3	7.2	7.5	0.0	0.0	Exact closure (0.0% residual)
Pelvis mesh	Pure triaxial ($x\mathbf{e}_x - \frac{1}{2}y\mathbf{e}_y - \frac{1}{2}z\mathbf{e}_z$)	84.8	11.1	4.1	0.0	0.0	Exact closure (0.0% residual)
Pelvis mesh	Pure bending ($-xy\mathbf{e}_x + \frac{1}{2}x^2\mathbf{e}_y$)	17.1	74.1	8.8	0.0	0.0	Exact closure (0.0% residual)
Pelvis mesh	Pure shear ($y\mathbf{e}_x + x\mathbf{e}_y$)	13.7	1.8	82.8	1.7	0.0	Exact closure (0.0% residual)
Pelvis mesh	Pure torsion ($yz\mathbf{e}_x - xy\mathbf{e}_z$)	5.5	0.5	7.0	87.1	0.0	Exact closure (0.0% residual)
Pelvis mesh	Infinitesimal rigid translation	0.0	0.0	0.0	0.0	0.0	Rigid: rejected ($f_{\text{nonrigid}} < 10^{-20}$)
Pelvis mesh	Infinitesimal rigid rotation	0.0	0.0	0.0	0.0	0.0	Rigid: rejected ($f_{\text{nonrigid}} < 10^{-20}$)
Pelvis mesh	Axial stretch + rigid perturbation	85.3	7.2	7.5	0.0	0.0	Exact closure (0.0% residual)



Fig. 11 Whole-pelvis deformation patterns of reference modes. Anterior (left) and cranial (right) views show displacement magnitude from purple to yellow, scaled to a 50 mm maximum; grey geometry is undeformed. Each bar partitions the whole-pelvis nonrigid squared shape norm into fitted axial tension/compression, bending, transverse shear, torsion, and residual. Infinitesimal rigid-body translation and rotation are projected out before the decomposition. The decomposition covers the complete pelvic ring in three-dimensional Cartesian coordinates. Overlapping pattern subspaces are allocated using Shapley values. “Fitted” is the sum of the four family shares of nonrigid squared displacement norm.

B Exploratory capacity–load associations

For each of the five most prevalent rank blocks, we calculated Spearman correlations across its cluster-member subjects between each single-functional max-nodal anatomical capacity $\kappa_{\ell,\infty}(Q_s)$ and its retained modal energy fraction f_C under each of the five prescribed loads. Figure 12 reports the correlation coefficients. These descriptive associations compare kinematic capacity with recruitment under the specified loads.

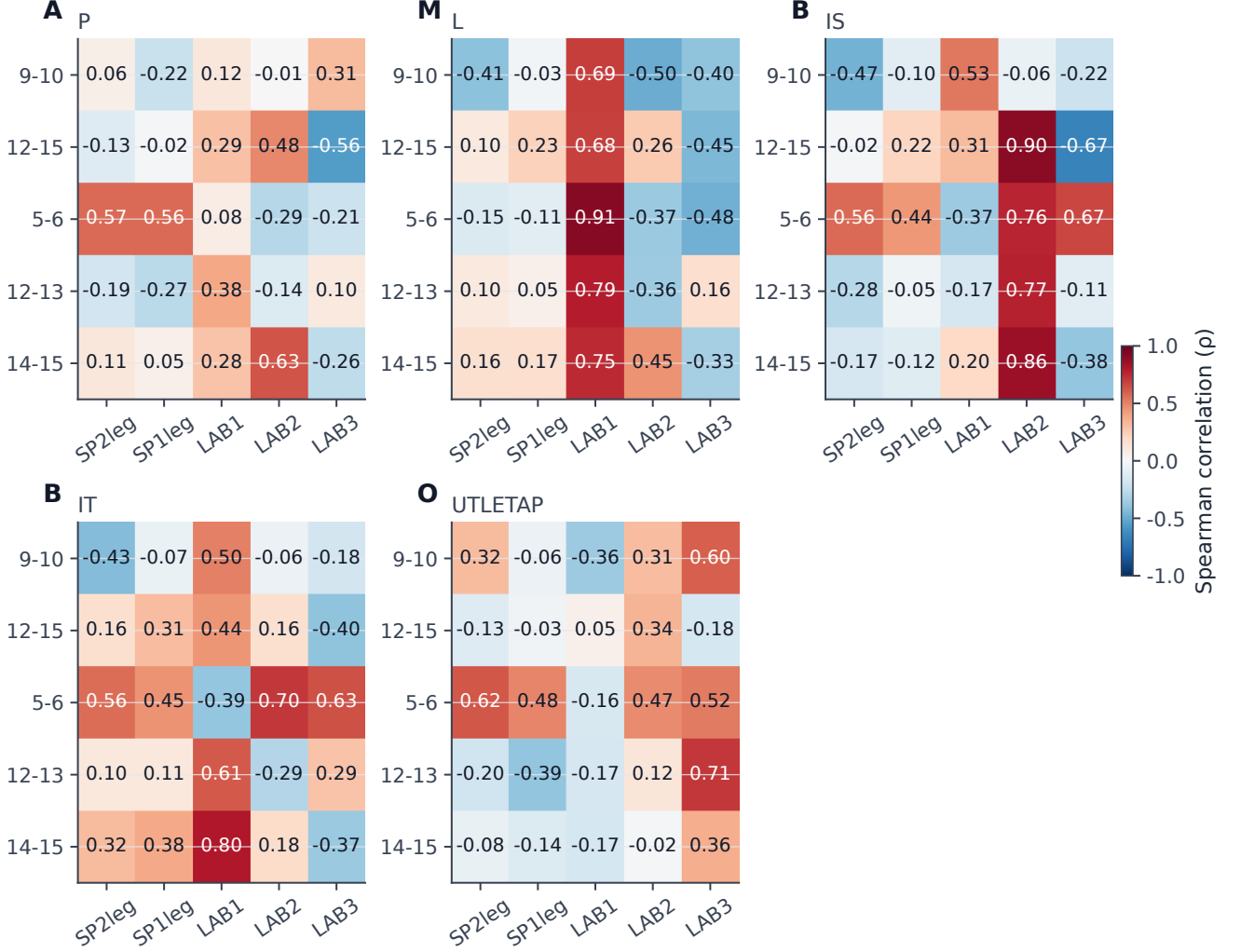


Fig. 12 Exploratory associations between anatomical capacity and load recruitment. Each cell gives the Spearman correlation between the indicated single-functional max-nodal capacity and the block’s retained modal energy fraction among subjects belonging to that rank block. Rows identify the five most prevalent blocks; columns identify the two standing and three comparative expansion loads. The panels correspond to AP and ML inlet, interspinous (BIS), bituberos (BIT), and AP outlet capacity.

C Exploratory allometry and eigenvalue variability

The post-hoc analysis relates archived eigenvalues to isotropic size, shape, material, and sex. Instrumental-variable regressions use volume, surface, and mass as size proxies; LMG/Shapley shares partition explained variance within the fitted model. These associations do not determine mode correspondence or anatomical capacity.

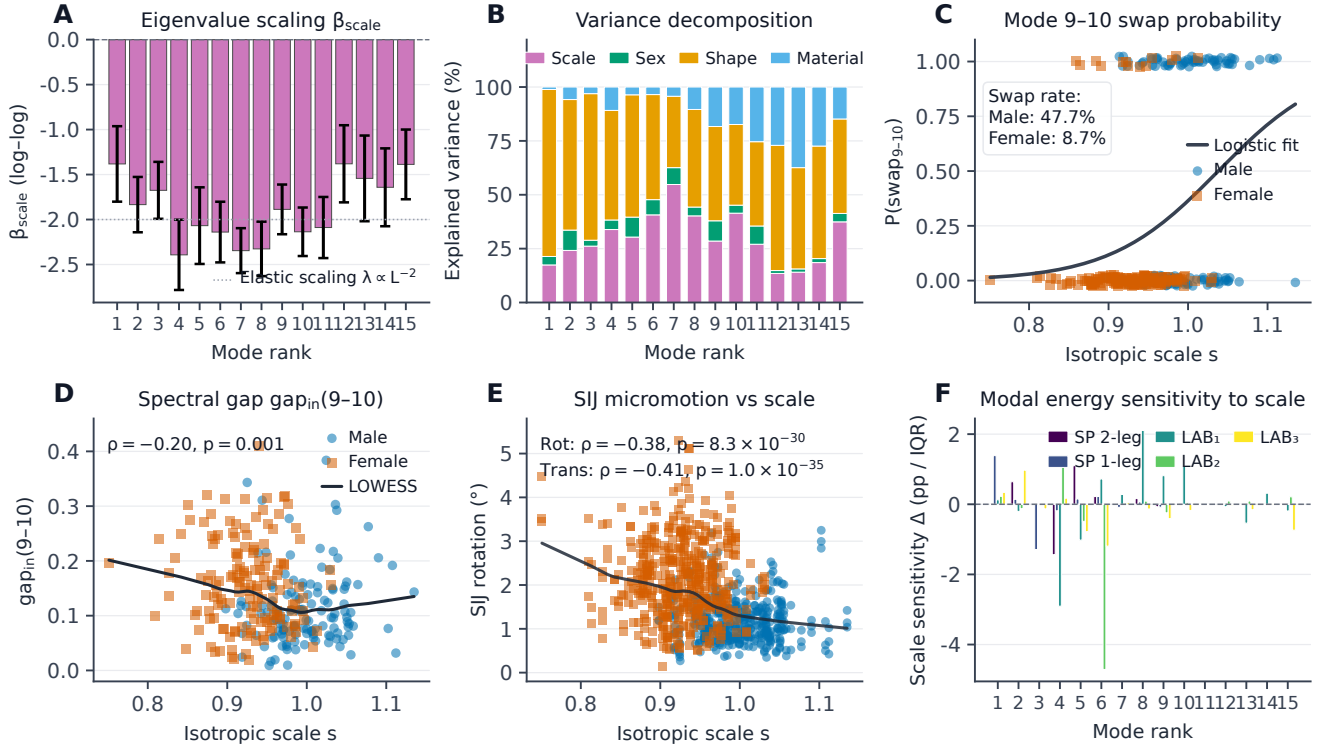


Fig. 13 Allometry dashboard

Table 8 Allometric scaling exponents and variance decomposition per eigenvalue mode. β_{scale} is the IV-2SLS log-log coefficient of eigenvalue on isotropic scale, with HC-robust 95 % CIs and FDR-corrected p -values. $\Delta\lambda$ per IQR gives the percent eigenvalue change for an interquartile range shift in log-scale. The last four columns show LMG (Shapley) shares of explained variance divided by full-model R^2 ; they sum to approximately 100% and are expressed relative to explained variance. The regressions describe associations in the archived cohort.

Mode	β_{scale}	95% CI	p_{FDR}	$\Delta\lambda$ IQR (%)	Scale share (%)	Shape share (%)	Material share (%)	Sex share (%)
1	-1.38	[-1.80, -0.96]	4.7e-10	-10.1	17.4	77.5	1.1	3.9
2	-1.83	[-2.14, -1.53]	8.4e-26	-13.1	24.2	60.6	5.8	9.4
3	-1.68	[-1.99, -1.36]	2.0e-21	-12.1	26.2	68.0	3.1	2.7
4	-2.39	[-2.78, -2.00]	7.7e-27	-16.8	33.9	50.8	10.9	4.4
5	-2.07	[-2.49, -1.64]	1.1e-18	-14.7	30.4	56.8	3.6	9.3
6	-2.14	[-2.48, -1.80]	3.4e-28	-15.1	40.7	48.8	3.5	7.1
7	-2.35	[-2.60, -2.10]	5.7e-49	-16.5	54.8	33.1	4.3	7.8
8	-2.33	[-2.63, -2.02]	2.4e-37	-16.4	40.2	45.3	10.4	4.1
9	-1.89	[-2.16, -1.61]	1.9e-31	-13.5	28.5	43.8	18.3	9.4
10	-2.14	[-2.41, -1.87]	7.7e-39	-15.1	41.5	37.4	17.4	3.7
11	-2.09	[-2.43, -1.75]	6.3e-27	-14.8	27.0	39.1	25.4	8.5
12	-1.38	[-1.81, -0.95]	9.4e-10	-10.1	13.5	58.1	27.1	1.3
13	-1.54	[-2.02, -1.07]	8.2e-10	-11.2	14.1	47.0	37.4	1.5
14	-1.64	[-2.07, -1.21]	1.5e-12	-11.8	18.5	52.2	27.4	1.9
15	-1.39	[-1.77, -1.00]	1.8e-11	-10.1	37.4	43.8	14.8	3.9

D First-order material-parameter sensitivity

D.1 Methods: first-order eigenvalue sensitivities and analytical uncertainty propagation

To propagate material-parameter uncertainty to the spectrum, we use classical first-order sensitivities of the generalised eigenproblem (10): for a parameter p_j that affects only the stiffness,

$$\frac{\partial \lambda_m}{\partial p_j} = \frac{\phi_m^\top (\partial K_s / \partial p_j) \phi_m}{\phi_m^\top M_s \phi_m}. \quad (46)$$

For ligament parameters, derivatives of local 3×3 blocks are $\partial W / \partial k_{\text{seg}} = \mathbf{e}\mathbf{e}^\top$ and $\partial W_g / \partial T_{\text{seg}} = (I - \mathbf{e}\mathbf{e}^\top) / L$. They enter the signed 6×6 segment assembly, with factor $1/N_{\text{seg}}$ for differentiation with respect to a bundle parameter. For a cartilage region with fixed Poisson ratio, $\partial K_s / \partial E_{\text{region}} = K_{\text{region}} / E_{\text{region}}$, where K_{region} is its

assembled baseline stiffness contribution. At source points on the high-density branch of (4), $\partial E/\partial\alpha = \rho_{\text{ash}}^\beta$ and $\partial E/\partial\beta = \alpha\rho_{\text{ash}}^\beta \ln \rho_{\text{ash}}$; the derivatives vanish on the constant branch. We interpolate each source derivative with the same radial-basis-function operator used for the modulus, then integrate it against the modal strain-energy density. This differentiates the implemented map, in which the nonlinear density law precedes interpolation. A finite-difference check at 1669 sampled cells in one subject agreed to relative L^2 errors below 4×10^{-11} for both parameters. For each positive material parameter p_j , let p_j^0 denote its baseline value and $\xi_j = \log(p_j/p_j^0)$. For each simple eigenvalue, we store the dimensionless logarithmic sensitivity

$$D_j^{(m)} = \frac{p_j^0}{\lambda_m^0} \frac{\partial \lambda_m}{\partial p_j} \Big|_{\mathbf{p}=\mathbf{p}^0} = \frac{\partial \log(\lambda_m/\lambda_m^0)}{\partial \xi_j} \Big|_{\xi=0}. \quad (47)$$

With $\mathbf{D}_m = (D_j^{(m)})_j$ and $\Sigma_\xi = \text{Cov}(\boldsymbol{\xi})$, first-order propagation gives $\text{Var}[\log(\lambda_m/\lambda_m^0)] \approx \mathbf{D}_m^\top \Sigma_\xi \mathbf{D}_m$ under the assumed material-parameter covariance. The implemented scenarios use log-space standard deviations (0.25, 0.20, 0.10) for global, type, and side effects of ligament stiffness; (0.35, 0.25, 0.12) for ligament pretension; and (0.35, 0.25, 0.10) for cartilage modulus. The global ligament stiffness–pretension correlation is 0.50. Bone-law parameters α and β have log-space standard deviations 0.30 and 0.20 with correlation -0.40 . These covariance scenarios were prescribed for the sensitivity analysis. Eigenvalue CoVs use the lognormal conversion $\sqrt{\exp(v) - 1}$ from propagated log-variance v . In this appendix, the superscript 0 denotes baseline-parameter eigenvalues, not the shared reference mass matrix M_{ref}^0 . For adjacent eigenvalues λ_i, λ_{i+1} , the variance of the gap $\Delta\lambda = \lambda_{i+1} - \lambda_i$ is computed from the combined sensitivity vector $\mathbf{g}_{i,i+1} = \lambda_{i+1}^0 \mathbf{D}_{i+1} - \lambda_i^0 \mathbf{D}_i$ as

$$\text{Var}(\lambda_{i+1} - \lambda_i) \approx \mathbf{g}_{i,i+1}^\top \Sigma_\xi \mathbf{g}_{i,i+1} \approx \text{Var}(\lambda_{i+1}) + \text{Var}(\lambda_i) - 2 \text{Cov}(\lambda_i, \lambda_{i+1}), \quad (48)$$

accounting for the substantial positive covariance induced by shared tissue parameters. The gap signal-to-noise ratio compares the nominal gap with its propagated standard deviation (Section D.2). It measures local susceptibility of modal ordering under the specified parameter covariance. The propagated quantities are eigenvalues and adjacent gaps. Sex-stratified results appear below.

D.2 Results: material sensitivity across spectral regions

First-order propagation with derivatives of the implemented material map predicts substantial variation in absolute eigenvalues under the assumed covariance; shared-parameter covariance reduces the calculated uncertainty of adjacent differences (Fig. 14; Table 9). Mean gap signal-to-noise ratios exceed 2 for ranks 1–2 and 2–3 in both sexes, whereas the 9–10 mean is 0.997 in females and 0.947 in males. Under this assumed local perturbation model, the internal 9–10 ordering is therefore more susceptible than the low-rank gaps. The calculation quantifies first-order uncertainty in eigenvalues and gaps under the assumed parameter covariance.

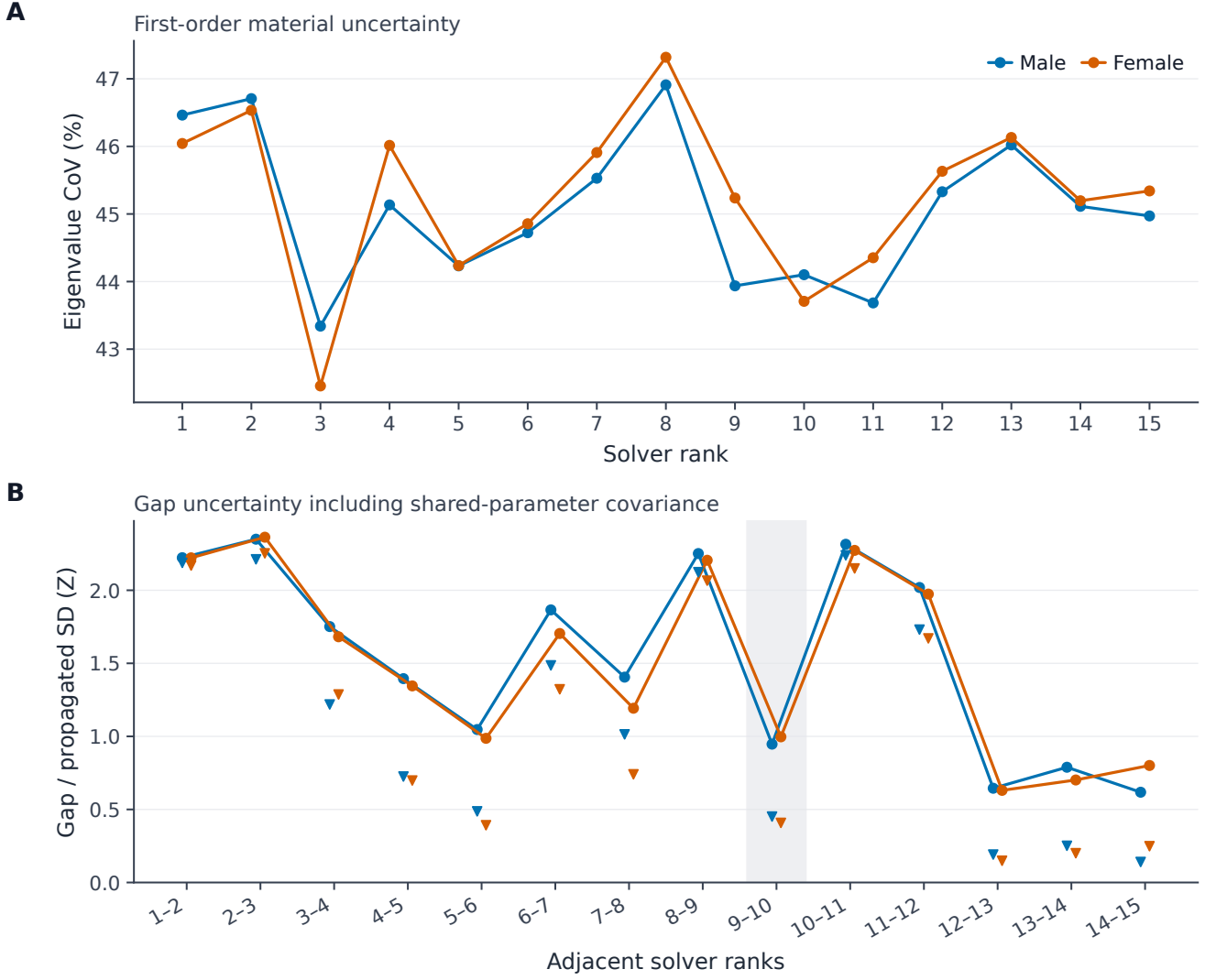


Fig. 14 Analytical propagation of material-parameter uncertainty to eigenvalues and spectral gaps. (A) Total coefficient of variation (CoV, %) of eigenvalues $\lambda_1, \dots, \lambda_{15}$ under hierarchical material-parameter covariance, stratified by sex (blue: male; yellow: female). (B) Relative gap signal-to-noise ratio (spectral gap divided by propagated standard deviation) between adjacent solver ranks, incorporating shared-parameter covariance $\text{Cov}(\lambda_i, \lambda_{i+1})$. The shaded region highlights the 9–10 rank transition. Triangles mark 10th-percentile values across subjects.

E Analytical example of close eigenvalues

E.1 Methods: analytical model for the near-degenerate regime

The cluster/subspace framework admits a compact analytical interpretation via a two-degree-of-freedom model. Let z_1, z_2 denote two generalised coordinates and define a stiffness matrix

$$K_2 = \begin{bmatrix} k_1 & c \\ c & k_2 \end{bmatrix}, \quad \Delta = k_1 - k_2, \quad k_1, k_2 > 0, \quad k_1 k_2 > c^2, \quad (49)$$

with detuning Δ and coupling c . The eigenvalues are

$$\lambda_{\pm} = \frac{k_1 + k_2}{2} \pm \sqrt{\left(\frac{\Delta}{2}\right)^2 + c^2}, \quad \tan(2\theta) = \frac{2c}{\Delta}, \quad (50)$$

and the minimum separation $\lambda_+ - \lambda_- = 2|c|$ is attained at $\Delta = 0$, for constant nonzero c as Δ is varied. With a continuous branch convention, $2\theta = \text{atan2}(2c, \Delta)$; at $\Delta = 0$ the angle is $\pi/4$ for $c > 0$, up to eigenvector sign. When $c = 0$, an exact crossing occurs. The operational criterion

$$|\Delta(\mu)| \leq \tau |c(\mu)| \quad \text{for some morphology parameter } \mu, \quad \tau \in [1.5, 2], \quad (51)$$

Table 9 Population-level material uncertainty propagation stratified by sex. CoV is the total coefficient of variation (ligament + cartilage + bone). Lig/Cart/Bone columns show each group's fraction of total variance. The bone contribution uses derivatives of the implemented density-to-modulus map. Gap signal-to-noise ratios appear in Fig. 14.

Mode	Female ($N = 150$)				Male ($N = 128$)			
	CoV (%)	Lig (%)	Cart (%)	Bone (%)	CoV (%)	Lig (%)	Cart (%)	Bone (%)
1	46.0	5	94	0	46.5	7	93	0
2	46.5	4	93	2	46.7	5	93	2
3	42.5	5	94	1	43.3	6	93	1
4	46.0	6	90	4	45.1	8	90	2
5	44.2	8	89	3	44.2	9	89	2
6	44.9	9	89	2	44.7	10	88	2
7	45.9	16	83	1	45.5	15	84	1
8	47.3	7	90	3	46.9	6	91	3
9	45.2	4	88	7	43.9	4	90	5
10	43.7	2	90	8	44.1	5	90	6
11	44.4	1	87	11	43.7	3	89	9
12	45.6	1	86	14	45.3	2	85	13
13	46.1	1	84	15	46.0	1	83	16
14	45.2	1	86	13	45.1	2	88	10
15	45.3	2	88	9	45.0	3	89	8

is an illustrative mixing criterion for the reduced model; it is distinct from the cohort cluster threshold in Eq. (12). Adding a third coordinate z_3 with stiffness k_3 and coupling vector $\mathbf{d} = [d_1, d_2]^\top$, dynamic reduction of the 3×3 eigenproblem $(K - \lambda I)\mathbf{z} = \mathbf{0}$ gives the exact non-linear reduced eigenproblem

$$\left(K_2 - \frac{\mathbf{d}\mathbf{d}^\top}{k_3 - \lambda}\right)\mathbf{z}_{12} = \lambda\mathbf{z}_{12}. \quad (52)$$

In the low-frequency limit ($\lambda \ll k_3$), static condensation yields the linear effective stiffness

$$K_{\text{eff}} = K_2 - \frac{\mathbf{d}\mathbf{d}^\top}{k_3} = \begin{bmatrix} k_1 - d_1^2/k_3 & c - d_1 d_2/k_3 \\ c - d_1 d_2/k_3 & k_2 - d_2^2/k_3 \end{bmatrix}, \quad (53)$$

showing how coupling to an additional coordinate shifts effective detuning and coupling. Depending on the parameters, the shift can move the system toward or away from avoided crossing. The assumption $\lambda \ll k_3$ excludes using this condensation for an arbitrarily soft hidden coordinate. All stochastic simulations solve the full 3×3 eigenproblem (Section E.2). The two-coordinate span is an analogue of an FE modal cluster, and the two-coordinate model relative gap is $(\lambda_+ - \lambda_-)/\lambda_-$. The illustrative eigenvector rotations are not identified with the anatomical-capacity optimisation in Eq. (19). A stochastic instantiation, evaluated in Section E.2, illustrates rank redistribution, paired projection, and load routing under the chosen model parameters. Section E.1.1 specifies the stochastic model used in the figure.

E.1.1 Stochastic implementation

The stochastic implementation evaluated in Section E.2 uses dimensionless illustrative parameters. The stiffness matrix is

$$K = \begin{bmatrix} 12 + \Delta/2 & c & d_1 \\ c & 12 - \Delta/2 & d_2 \\ d_1 & d_2 & k_3 \end{bmatrix}, \quad M = I. \quad (54)$$

For morphology coordinate μ and nuisance scale η , independent standard-normal draws determine

$$\Delta = 1.6\mu + (0.05 + 0.22\eta)Z_\Delta, \quad (55)$$

$$c = |0.22 + (0.008 + 0.018\eta)Z_c|, \quad (56)$$

$$k_3 = 5.2 \exp[(0.02 + 0.12\eta)Z_3], \quad (57)$$

$$d_1 = 0.28 + 0.42\eta + (0.01 + 0.08\eta)Z_1, \quad (58)$$

$$d_2 = 0.24 + 0.40\eta + (0.01 + 0.08\eta)Z_2. \quad (59)$$

Draws are accepted only when all eigenvalues exceed 10^{-10} and the largest absolute component of the third coordinate occurs in the lowest eigenvector. This acceptance criterion retains a low-rank backbone by construction. Up to 200 draws are attempted per subject.

The anatomical observable and task-like load are unit vectors proportional to $(1, 0, 0.2)^\top$; the standing-like load is a unit vector proportional to $(0.10, 0.05, 1)^\top$. Writing $\mathbf{b}$ for the unit observable and $\mathbf{f}$ for a unit load, the observable projection of a unit eigenvector $\mathbf{v}_i$ is $(\mathbf{v}_i^\top \mathbf{b})^2$ and the upper-pair score is the sum for $i = 2, 3$. Load fractions are proportional to $(\mathbf{v}_i^\top \mathbf{f})^2 / \lambda_i$. The analytical example uses squared projections as illustrative observables; the pelvic cohort uses max-nodal anatomical capacity.

The main illustrative cohort contains 240 evenly spaced coordinates in $[-1.10, 1.10]$, perturbed by normal noise of standard deviation 0.035 (random seed 11), sorted, and simulated at $\eta = 0.85$ (seed 19). The nuisance sweep uses nine values in $[0, 1.6]$ and 480 coordinates in $[-0.35, 0.35]$, with seed 31 at each value. Carrier change is the proportion whose larger upper-pair observable projection changes rank relative to $\eta = 0$. Pair projection and task routing are summarised by their medians.

For every accepted draw, the full 3×3 eigenproblem gives the modal ordering and observable projections. The static-condensation formula in Eq. (53) provides the limiting interpretation for $\lambda \ll k_3$.

E.2 Results: label–subspace dissociation

The constructed analytical example illustrates how a separated low-rank mode and a coupled higher pair can produce contrasting rank and subspace behaviour (Fig. 15). As detuning varies, the dominant projection of the anatomical observable transfers between the upper eigenvectors, whereas their summed projection remains high. Standing-like and task-like loads recruit different spectral regions. Increasing the nuisance scale changes the dominant carrier more strongly than the upper-pair projection or task routing.

The analytical example conditions accepted draws on a retained low-rank mode and measures coupling by squared projection.

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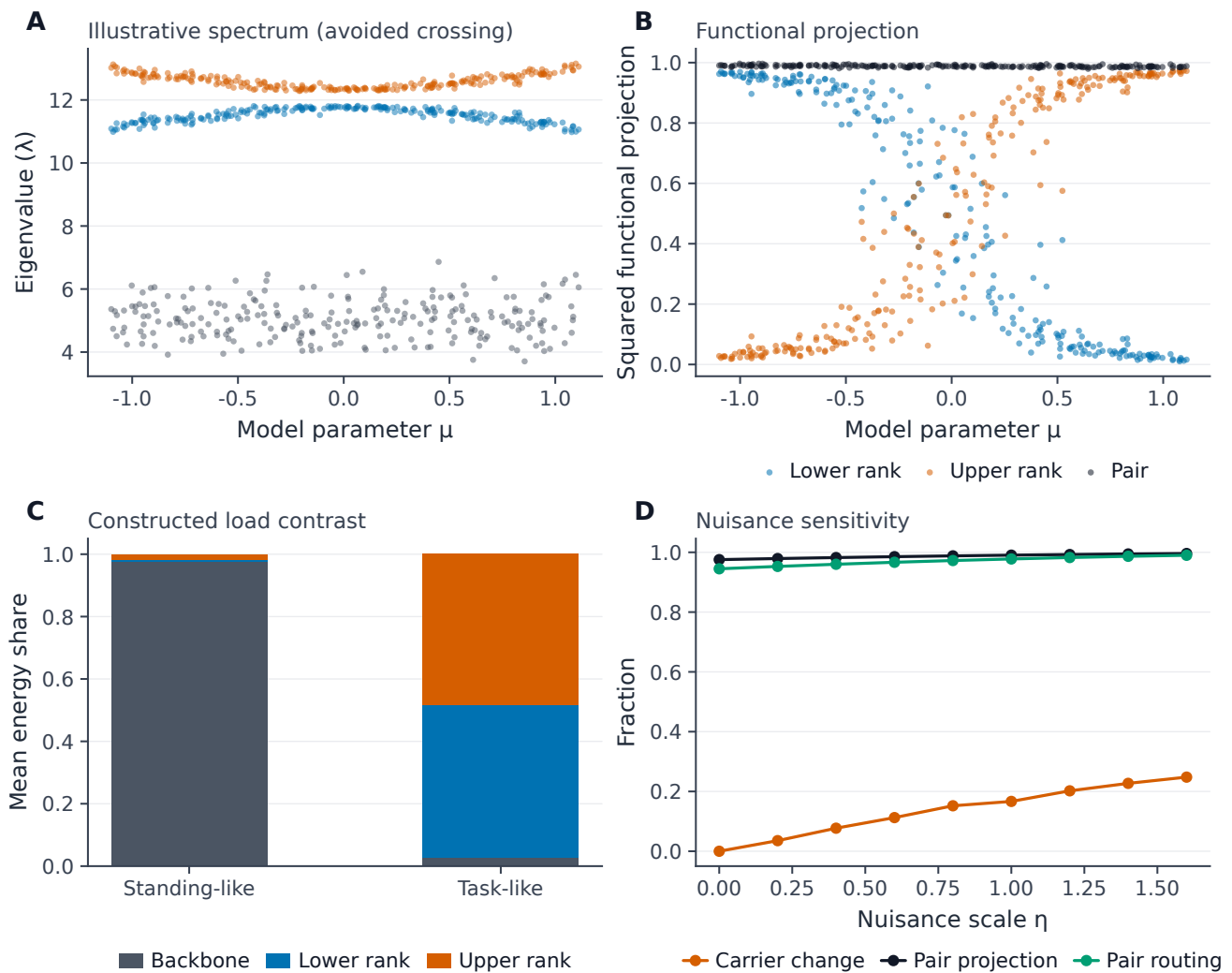


Fig. 15 Minimal analytical model illustrates label exchange within a stable subspace. A stochastic cohort of the analytical model with morphology parameter μ and hidden-pathway scale η . **(A)** Spectrum: a separated backbone mode and a nearby eigenvalue pair undergoing near-degeneracy around $\mu \approx 0$. **(B)** Functional projection: coupling shifts between individual rank labels across the transition zone while total coupling within the pair remains high. **(C)** Load routing: standing-like load concentrates in the backbone, whereas task-like load recruits the clustered block with a variable between-label split. **(D)** Effect of hidden-pathway variability scale η : label-exchange rate increases with η , whereas subspace coupling and task routing remain stable. Model derivations and simulation parameters are detailed in Sections E.1–E.1.1.

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